01-(MCQ,Marks-01)
$\frac{x}{A}+\frac{y}{B}=1$, is a linear equation of a straight line in ——- form.
Intercept *(correct)*
point slope
two points
normal

02-(MCQ,Marks-01)
Equation of $x-$axis is—–.
$x=0$
$y=0$*(correct)*
$x=a$
$y=b$

03-(MCQ,Marks-01)
Equation of $y-$axis is—–.
$x=a$
$y=b$
$x=0$*(correct)*
$y=0$

04-(MCQ,Marks-01)
Equation of line parallel to x-axis is——-.
$x=0$
$y=0$
$x=a$
$y=b$*(correct)*

05-(MCQ,Marks-01)
Equation of line parallel to y-axis is——-.
$x=0$
$y=0$
$x=a$*(correct)*
$y=b$

06-(MCQ,Marks-01)
Equation of a straight line bisecting 1st and 3rd quadrants is——-.
$y=x$*(correct)*
$y=-x$
$y=x+1$
$y=-x+1$

07-(MCQ,Marks-01)
Equation of a straight line bisecting 2nd and 4th quadrants is——-.
$y=x$
$y=-x$*(correct)*

$y = x + 1$
$y = -x + 1$

08-(MCQ,Marks-01)
$f(x) = ax^2 + bx + c$,graphically represents ——.
a parabola if $a \neq 0$
a straight line if $a = 0$
all above are true *(correct)*

09-(MCQ,Marks-01)
For the quadratic equation:$ax^2 + bx + c = 0$, roots will be real and distinct if the Discriminant= $\sqrt{b^2 - 4ac} - - - -$.
> 0*(correct)*
< 0
$= 0$
$\neq 0$

10-(MCQ,Marks-01)
For the quadratic equation:$ax^2 + bx + c = 0$, roots will be equal if the Discriminant= $\sqrt{b^2 - 4ac} - - - -$.
> 0
< 0
$= 0$*(correct)*
$\neq 0$

11-(MCQ,Marks-01)
For the quadratic equation:$ax^2 + bx + c = 0$, roots will be irrational and unequal if the Discriminant= $\sqrt{b^2 - 4ac} - - - -$.
is a perfect square
is not perfect square *(correct)*

12-(MCQ,Marks-01)
For the quadratic equation:$ax^2 + bx + c = 0$, roots will be rational if the Discriminant= $\sqrt{b^2 - 4ac} - - - -$.
is a perfect square *(correct)*
is not perfect square

13-(MCQ,Marks-01)
For the quadratic equation:$ax^2+bx+c = 0$, roots will be complex conjugates of each other, if the Discriminant $= \sqrt{b^2 - 4ac} - - - -$.
> 0
< 0*(correct)*
$= 0$
$\neq 0$

14-(MCQ,Marks-01)

If α, β are the roots of a quadratic equation $ax^2 + bx + c = 0, a \neq 0$, then sum of roots= $\alpha + \beta =$——.

$\frac{-b}{a}$ *(correct)*

$\frac{b}{a}$

$\frac{c}{a}$

$\frac{-c}{a}$

15-(MCQ,Marks-01)

If α, β are the roots of a quadratic equation $ax^2 + bx + c = 0, a \neq 0$, then the product of roots= $\alpha\beta =$——.

$\frac{-b}{a}$

$\frac{b}{a}$

$\frac{c}{a}$ *(correct)*

$\frac{-c}{a}$

16-(MCQ,Marks-01)

If α, β are the given roots of a required quadratic equation, such that $S = \alpha + \beta$ and $P = \alpha\beta$, then the quadratic equation is given by——-.

$x^2 + Sx + P = 0$

$x^2 - Sx + P = 0$ *(correct)*

$x^2 + Sx - P = 0$

$x^2 - Sx - P = 0$

17-(MCQ,Marks-01)

The cubic equation $ax^3 + bx^2 + cx + d = 0$ will have three roots, if

$d \neq 0$

$c \neq 0$

$b \neq 0$

$a \neq 0$ *(correct)*

18-(MCQ,Marks-01)

If α, β and γ are the given roots of a cubic equation:$ax^3 + bx^2 + cx + d, a \neq 0$, then the Sum of roots= $S = \alpha + \beta + \gamma =$——-.

$\frac{c}{a}$

$\frac{d}{a}$

$\frac{-c}{a}$

$\frac{-b}{a}$ *(correct)*

19-(MCQ,Marks-01)

If α, β and γ are the given roots of a cubic equation:$ax^3 + bx^2 + cx + d, a \neq 0$, then the Sum of product of roots= $SoP = \alpha\beta + \beta\gamma + \alpha\gamma =$——-.

$\frac{c}{a}$ *(correct)*

$\frac{d}{a}$

$\frac{-c}{a}$

$\frac{-b}{a}$

20-(MCQ,Marks-01)

If α, β and γ are the given roots of a cubic equation:$ax^3 + bx^2 + cx + d, a \neq 0$, then the product of roots= $P = \alpha\beta\gamma =$——-.

$\frac{c}{a}$
$\frac{d}{a}$
$\frac{-d}{a}$ *(correct)*
$\frac{-b}{a}$

21-(MCQ,Marks-01)

If α, β and γ are the given roots of a required cubic equation, such that:
Sum of product of roots $= S\text{o}P = \alpha\beta + \beta\gamma + \alpha\gamma$ and
Product of roots $= P = \alpha\beta\gamma$, then the cubic equation is given by——-.

$x^3 + Sx^2 + (S\text{o}P)\, x + P = 0$
$x^3 - Sx^2 + (S\text{o}P)\, x + P = 0$
$x^3 + Sx^2 - (S\text{o}P)\, x + P = 0$
$x^3 - Sx^2 + (S\text{o}P)\, x - P = 0$ *(correct)*

22-(MCQ,Marks-01)

If $\frac{dy}{dx} = y$, then which of the following would be an appropriate implication?

$y = A + e^x$
$y = A - e^x$
$y = Ae^x$ *(correct)*
$y = \frac{A}{e^x}$

23-(MCQ,Marks-01)

Which of the following is an example of Differential Equation?

$x^2 + y^2 = 1$
$y = mx + c$
$\frac{dy}{dx} = y$*(correct)*
$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = k$

24-(MCQ,Marks-01))

Which of the following is an example of Partial Differential Equation?

$x^2 + y^2 = 1$
$y = mx + c$
$\frac{dy}{dx} = y$
$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = k$*(correct)*

01-MCQ, Marks-01

Which of the following will represent an ordinary differential equation?

$\frac{d^2y}{dx^2} - 5\left(\frac{dy}{dx}\right)^2 = 0$ (*Correct*)

$\frac{d}{dx}(\sin x) = \cos x$

$\frac{d}{dx}(\ln x) = \frac{1}{x}$

$\frac{d}{dx}\left(\frac{y}{x}\right) = \frac{x\frac{dy}{dx} - y}{x^2}$

02-MCQ, Marks-01

For a differential equation, the order of highest derivatives appearing in it, is called the ——- of differential equation.

degree

order (*Correct*)

exponent

power

03-MCQ, Marks-01

The ordinary differential equation: $\frac{d^4y}{dx^4} - 2\left(\frac{dy}{dx}\right)^5 + 3xy = \sin x$ is of order——.

6

5

4 (*Correct*)

1

04-MCQ, Marks-01

The ordinary differential equation: $\left(\frac{d^3y}{dx^3}\right)\left(\frac{d^2y}{dx^2}\right) + 6\frac{dy}{dx} = \frac{1}{x}$ is of order——.

6

5

4

3 (*Correct*)

05-MCQ, Marks-01

The ordinary differential equation: $y' = 4y + x^3$ is of order——.

1(*Correct*)

2

3

4

06-MCQ, Marks-01

The ordinary differential equation: $(2x+y)dx + (y+3x)dy = 0$ is of order——.

1(*Correct*)

2

3

4

07-MCQ, Marks-01
The ordinary differential equation: $4y'' + \cos x . y' - y = e^{\tan x}$ is of order——.
1
2 (*Correct*)
3
4

08-MCQ, Marks-01
The ordinary differential equation: $(y')^3 = \frac{a \sec x}{b \ln x}$ is of order——.
1(*Correct*)
2
3
4

09-MCQ, Marks-01
The ordinary differential equation: $\left(5\frac{dy}{dx}\right)^3 = \sqrt{2 - \left(3\frac{dy}{dx}\right)^2}$ is of order——.
1 (*Correct*)
2
3
6

10-MCQ, Marks-01
The ordinary differential equation: $\sqrt[3]{\left(5\frac{d^2y}{dx^2}\right)^2} = \sqrt{2 - \left(3\frac{dy}{dx}\right)^2}$ is of order——.
6
5
4
2 (*Correct*)

11-MCQ, Marks-01
The ordinary differential equation: $\sqrt[5]{5\frac{dy}{dx}} = \frac{d^3y}{dx^3}$ is of order——.
2
3(*Correct*)
5
6

12-MCQ, Marks-01
The ordinary differential equation: $8\frac{d^2x}{dy^2} + 5x = \csc 2y$ is of order——.
1
2 (*Correct*)
3
4

13-MCQ, Marks-01

The ordinary differential equation: $7\frac{d^2y}{dx^2} + 2x^2y = \cot\left(\frac{d^2y}{dx^2}\right)$ is of order——.
undefined
4
1
2 (*Correct*)

14-MCQ, Marks-01

The ordinary differential equation:$x = 1+xy\left(\frac{dy}{dx}\right)+\frac{x^2y^2}{2}\left(\frac{dy}{dx}\right)^2+\frac{x^3y^3}{6}\left(\frac{dy}{dx}\right)^3\cdots$ is of order——.
1(*Correct*)
2
3
undefined

15-MCQ, Marks-01

The ordinary differential equation: $t = 1 - \frac{1}{2!}\left(\frac{dx}{dt}\right)^2 + \frac{1}{4!}\left(\frac{dx}{dt}\right)^4 - \frac{1}{6!}\left(\frac{dx}{dt}\right)^6 + \cdots\cdots$ is of order——.
1(*Correct*)
2
4
6

16-MCQ, Marks-01

A differential equation which can be rationalized and cleared of fractions with respect to all its derivatives present, then the exponent(power) of the highest order derivatives is called the ——- of a differential equation.
order
degree (*Correct*)
linearity
homogeneity

17-MCQ, Marks-01

The differential equation:$(3y''')^{\frac{2}{3}} = 6 + y'$ is of degree——.
$\frac{2}{3}$
3
2 (*Correct*)
1

18-MCQ, Marks-01

The differential equation:$3y''' = 6x - 5y$ is of degree——.
$\frac{2}{3}$
3
2
1(*Correct*)

19-MCQ, Marks-01

The degree of a differential equation ——— if the unknown function(dependant variable) is an argument(so called the angle) of the transcendental functions.

is undefined (*Correct*)

can be defined

is unique

is stable

20-MCQ, Marks-01

The degree of a differential equation ——— if the unknown function(dependant variable) y can not be written as polynomial in y and its derivatives.

is undefined (*Correct*)

can be defined

is unique

is stable

21-MCQ, Marks-01

The degree of differential equation: $x^2y''' + \cos x.y'' - \sin(xy) = 1$ is ——-.

2

3

1

undefined (*Correct*)

22-MCQ, Marks-01

The degree of differential equation: $x^2y'' + 3(y')^2 = x \ln y''$ is ——-.

2

3

1

undefined (*Correct*)

23-MCQ, Marks-01

The degree of differential equation: $\sec(y') = y' - 2x + 1$ is ——-.

2

3

1

undefined (*Correct*)

24-MCQ, Marks-01

The ordinary differential equation: $\frac{d^4y}{dx^4} - 2\left(\frac{dy}{dx}\right)^5 + 3xy = \sin x$ is of degree——.

6

5

4

1 (*Correct*)

25-MCQ, Marks-01

The ordinary differential equation: $\left(\frac{d^3y}{dx^3}\right)\left(\frac{d^2y}{dx^2}\right)+6\frac{dy}{dx}=\frac{1}{x}$ is of degree——.
1(*Correct*)
5
4
3

26-MCQ, Marks-01
The ordinary differential equation: $y'=4y+x^3$ is of degree——.
1(*Correct*)
2
3
4

27-MCQ, Marks-01
The ordinary differential equation: $(2x+y)dx+(y+3x)dy=0$ is of degree——.
1(*Correct*)
2
3
4
28-MCQ, Marks-01
The ordinary differential equation: $4y''+\cos x.y'-y=e^{\tan x}$ is of degree——.
1(*Correct*)
2
3
4

29-MCQ, Marks-01
The ordinary differential equation: $(y')^3=\frac{a\sec x}{b\ln x}$ is of degree——.
1
2
3(*Correct*)
4

30-MCQ, Marks-01
The ordinary differential equation: $\left(5\frac{dy}{dx}\right)^3=\sqrt{2-\left(3\frac{dy}{dx}\right)^2}$ is of degree——.
1
2
3
6 (*Correct*)

31-MCQ, Marks-01
The ordinary differential equation: $\sqrt[3]{\left(5\frac{d^2y}{dx^2}\right)^2}=\sqrt{2-\left(3\frac{dy}{dx}\right)^2}$ is of degree——.

6
5
4($Correct$)
2

32-MCQ, Marks-01

The ordinary differential equation: $\sqrt[5]{5\frac{dy}{dx}} = \frac{d^3y}{dx^3}$ is of degree——.
2
3
5($Correct$)
6

33-MCQ, Marks-01

The ordinary differential equation: $8\frac{d^2x}{dy^2} + 5x = \csc 2y$ is of degree——.
1($Correct$)
2
3
4

34-MCQ, Marks-01

The ordinary differential equation: $7\frac{d^2y}{dx^2} + 2x^2y = \cot\left(\frac{d^2y}{dx^2}\right)$ is of degree——.
undefined($Correct$)
4
1
2

35-MCQ, Marks-01

The ordinary differential equation:$x = 1 + xy\left(\frac{dy}{dx}\right) + \frac{x^2y^2}{2}\left(\frac{dy}{dx}\right)^2 + \frac{x^3y^3}{6}\left(\frac{dy}{dx}\right)^3 \cdots$ is of degree——.
1($Correct$)
2
3
undefined

36-MCQ, Marks-01

The ordinary differential equation: $t = 1 - \frac{1}{2!}\left(\frac{dx}{dt}\right)^2 + \frac{1}{4!}\left(\frac{dx}{dt}\right)^4 - \frac{1}{6!}\left(\frac{dx}{dt}\right)^6 + \cdots\cdots$ is of order——.
1($Correct$)
2
4
6

37-MCQ, Marks-01

Given a equation of family of curves, we can form the corresponding differential equation by——— and then —– the arbitrary constants.
integrating,eliminating
integrating,including
differentiating,eliminating (*Correct*)
differentiating, including

38-MCQ, Marks-01
An equation involving n-arbitrary constants will give rise to a differential equation of ——- order.

$(n+2)^{th}$
$(n+1)^{th}$

n^{th}(*Correct*)
$(n-1)^{th}$

39-MCQ, Marks-01
The solution of a differential equation in which the number of independent arbitrary constants is the same as the order of the differential equation is called——- solution.
trivial
singular
particular
general (*Correct*)

40-MCQ, Marks-01
The solution of a differential equation which can't be obtained from the general solution by any choice of the independent arbitrary constant is called the ——— solution.
trivial
singular (*Correct*)
particular
complementary

41-MCQ, Marks-01
In practice, we are usually assigned to determine the solution of a differential equation which satisfies certain prescribed side conditions. If the conditions are specified at one point such as $y(x_0) = A$ and $y'(x_0) = B$, then the given problem is called ————.
initial value problem(IVP) (*Correct*)
boundary value problem(BVP)

42-MCQ, Marks-01

In practice, we are usually assigned to determine the solution of a differential equation which satisfies certain prescribed side conditions. If the conditions are specified at more than one point such as $y(x_0) = A$ and $y'(x_1) = B$, then the given problem is called ————.

initial value problem(IVP)

boundary value problem(BVP) (*Correct*)

43-MCQ, Marks-01

Which of the following is an example of IVP?

$y'' - y' - 12 = 0, \;\; y(0) = -2, y'(0) = 6$ (*Correct*)

$y'' - y' - 12 = 0, \;\; y(0) = -2, y'(1) = 6$

$y'' - y' - 12 = 0$

$y'' - y' - 12 = x$

44-MCQ, Marks-01

Which of the following is an example of BVP?

$y'' - y' - 12 = 0, \;\; y(0) = -2, y'(0) = 6$

$y'' - y' - 12 = 0, \;\; y(0) = -2, y'(1) = 6$ (*Correct*)

$y'' - y' - 12 = 0$

$y'' - y' - 12 = x$

MCQ, Marks-01

The constant solution of the differential equation: $\frac{dy}{dx} = \frac{\sin y}{e^x}$ is ——.

$y = 2n\pi$

$y = n\frac{\pi}{2}$

$y = n\pi$ (*correct*)

$y = (n+1)\pi$, where $n \in \mathbb{Z}$

Descriptive, Marks-02

Express the differential equation: $k\left(y^2 + y'\right) = y - xy'$ into separable form.

Solution:

$k\left(y^2 + y'\right) = y - xy' \Longrightarrow ky' + xy' = y - ky^2 \Longrightarrow y'\left(k + x\right) = y\left(1 - ky\right)$

$\frac{dy}{dx}\left(k + x\right) = y\left(1 - ky\right)$

Now separating the variables;

$\boxed{\frac{dy}{dx} = \frac{1}{(k+x)}\left(y - ky^2\right)}$ i.e. the required separable form.

Descriptive, Marks-03

Express the differential equation: $x^2 - yy' = e^x$ into separable form. Also find its constant solution.

Solution:

$x^2 - e^y y' = e^x \Longrightarrow x^2 - e^x = e^y\frac{dy}{dx} \Longrightarrow \boxed{\frac{dy}{dx} = \left(x^2 - e^x\right)e^{-y} = f(x).g(y)}$ i.e. the required separable form, where

$f(x) = x^2 - e^x$ and $g(y) = e^{-y}$. For the constant solution, put $g(y) = e^{-y} = 0 \Longrightarrow \frac{1}{e^y} = 0$

$\Longrightarrow e^y \longrightarrow \infty \Longrightarrow \boxed{y \longrightarrow \infty} \Longrightarrow \nexists$ a real valued constant solution.

Descriptive, Marks-05

Determine the particular solution of the initial value problem:

$x\cos x dx + \left(1 - 6y^5\right)dy = 0$, $y(\pi) = 0$.

Solution:

$x\cos x dx + \left(1 - 6y^5\right)dy = 0$

Separating the variables;

$\left(1 - 6y^5\right)dy = -x\cos x dx$

Integrating;

$\int\left(1 - 6y^5\right)dy = -\int x\cos x dx - - - (1)$

$$\left\{\begin{array}{c} \text{For } \int x\cos x dx \\ \text{Integrating by parts by taking } x \text{ as first function and } \cos x \text{ as 2nd one.} \\ \int x\cos x dx = x\left(\int\cos x dx\right) - \int\left(\left(\int\cos x dx\right)\frac{d}{dx}x\right)dx \\ \Longrightarrow \int x\cos x dx = x\sin x - \int\left(\sin x.1\right)dx \\ \Longrightarrow \int x\cos x dx = x\sin x - \left(-\cos x\right) = \cos x + x\sin x \end{array}\right.$$

$\therefore(1) \Longrightarrow y - y^6 = -\left(\cos x + x\sin x\right) + c$

For c, put $x = \pi$ and $y = 0$ in above;

$0 - 0 = -\left(\cos\pi + \pi\sin\pi\right) + c \Longrightarrow 0 = -(-1 + 0) + c \Longrightarrow c = -1$

$\therefore$the particular solution is; $\boxed{y - y^6 = -\left(\cos x + x\sin x\right) - 1}$

MCQ, Marks-01

Which of the following will represent a homogeneous differential equation?

$xy' = y + xe^{\frac{y}{x}}$

$y(ydx - xdy) + x\sqrt{x^2+y^2}dy = 0$

$\left(x\tan\frac{y}{x} - y\sec^2\frac{y}{x}\right)dx + x\sec^2\frac{y}{x}dy = 0$

All of the above (*correct*)

Descriptive, Marks-02

Reason out properly how does the function:$f(x,y) = \frac{1+e^{\frac{x}{y}}}{e^{\frac{x}{y}}\left(\frac{x}{y}-1\right)}$ is homogeneous?

Solution:

A multivaraible function $f(x,y)$ would be homogeneous of degree $n \in \mathbb{R}$ if and only if

$f(tx,ty) = t^n f(x,y)$

$\therefore$ for the given, taking:

$f(tx,ty) = \frac{1+e^{\frac{tx}{ty}}}{e^{\frac{tx}{ty}}\left(\frac{tx}{ty}-1\right)} = 1.\frac{1+e^{\frac{x}{y}}}{e^{\frac{x}{y}}\left(\frac{x}{y}-1\right)} = t^0 f(x,y) \Longrightarrow$given $f(x,y)$ is homogeneous function of degree *zero*.

Descriptive, Marks-03

If $y' = \frac{x-y-1}{x+3y+5}$,then by using the substitutions $x = X+h$ and $y = Y+k$, find the values of h and k for whom the given DE is homogeneous.

Solution:

If $x = X+h$ and $y = Y+k \Longrightarrow y' = \frac{dy}{dx} = \frac{d(Y+k)}{d(X+h)} = \frac{dY}{dX}$

$\therefore$given DE becomes,

$\frac{dY}{dX} = \frac{(X+h)-(Y+k)-1}{(X+h)+3(Y+k)+5} = \frac{X-Y+(h-k-1)}{X+3Y+(h+3k+5)} \Longrightarrow$ the given DE will be homogeneous$\Longleftrightarrow$

$$\begin{cases} h-k-1=0 \\ h+3k+5=0 \end{cases}$$

$$\Longrightarrow \begin{cases} h-k=1 \\ h+3k=-5 \end{cases}$$

By subtracting; $-4k = 6 \Longrightarrow \boxed{k = -\frac{3}{2}}$, put this in 1st to get h.

$\Longrightarrow h + \frac{3}{2} = 1 \Longrightarrow \boxed{h = -\frac{1}{2}}$

Descriptive, Marks-05

Transform the following homogeneous differential equation into separable form by using the appropriate substitution;

$y(ydx - xdy) + x\sqrt{x^2+y^2}dy = 0$

Solution:

$y(ydx - xdy) + x\sqrt{x^2+y^2}dy = 0$

$\Longrightarrow y^2dx - xydy + x\sqrt{x^2+y^2}dy = 0 \Longrightarrow y^2dx = xydy - x\sqrt{x^2+y^2}dy$

$\Longrightarrow y^2dx = \left(xy - x\sqrt{x^2+y^2}\right)dy \Longrightarrow \frac{dy}{dx} = \frac{y^2}{\left(xy-x\sqrt{x^2+y^2}\right)}$——(1)

Put $y = vx \Longrightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore$(1) becomes;

$v + x\frac{dv}{dx} = \frac{v^2x^2}{\left(xvx - x\sqrt{x^2+v^2x^2}\right)} = \frac{v^2x^2}{\left(vx^2 - x^2\sqrt{v^2+1}\right)} = \frac{v^2}{v-\sqrt{v^2+1}}$

$\implies x\frac{dv}{dx} = \frac{v^2}{v-\sqrt{v^2+1}} - v = \frac{v^2 - v\left(v-\sqrt{v^2+1}\right)}{v-\sqrt{v^2+1}} = \frac{v^2+v\sqrt{v^2+1}-v^2}{v-\sqrt{v^2+1}} = \frac{v\sqrt{v^2+1}}{v-\sqrt{v^2+1}}$

$\implies \boxed{\frac{dv}{dx} = \frac{1}{x}\left(\frac{v\sqrt{v^2+1}}{v-\sqrt{v^2+1}}\right) = f(x).g(v)}$, i.e. the required separable form, where $f(x) = \frac{1}{x}$ and $g(v) = \left(\frac{v\sqrt{v^2+1}}{v-\sqrt{v^2+1}}\right)$

Lecture 5

Question-01 (MCQ, Marks-01, Expected completion time= 1 min.)

If $\frac{dy}{dx} = \frac{ax^2-by}{bx+ay^2}$, then which of the following is the most accurate option? It is a(an) ———.

Exact differential equation.

Separable differential equation.

Homogeneous differential equation.

Homogeneous as well as an Exact differential equation. (*Correct*)

Question-02 (Descriptive, Marks-02,Expected completion time= 4 mins.)

If $(y\cos x + 2xe^y)\,dx + \left(\sin x + x^2e^y - 1\right)dy = 0$ is an Exact differential equation of the form say; $d(F(x,y)) = 0$, then find $\frac{\partial F}{\partial x}$ and $\frac{\partial F}{\partial y}$?

Solution:

From differential Calculus, $d(F(x,y)) = \frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial y}dy$

$\therefore d(F(x,y)) = \frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial y}dy = 0$

But we have given, $(y\cos x + 2xe^y)\,dx + \left(\sin x + x^2e^y - 1\right)dy = 0$ is an Exact differential equation.

$\therefore$ Comparing the coefficients of dx and dy $\Longrightarrow \boxed{\frac{\partial F}{\partial x} = y\cos x + 2xe^y}$ and $\boxed{\frac{\partial F}{\partial y} = \sin x + x^2e^y - 1}$.

Question-03 (Descriptive, Marks-03, Expected completion time= 6 min.)

Determine whether the differential equation;$(y\cos x + 2xe^y)\,dx + \left(\sin x + x^2e^y - 1\right)dy = 0$ is Exact or not?

Solution:

$(y\cos x + 2xe^y)\,dx + \left(\sin x + x^2e^y - 1\right)dy = 0$

Comparing with the general form:$M(x,y)dx + N(x,y)dy = 0$

$\Longrightarrow M(x,y) = y\cos x + 2xe^y$ and $N(x,y) = \sin x + x^2e^y - 1$

Now $\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(y\cos x + 2xe^y) = \frac{\partial}{\partial y}(y\cos x) + \frac{\partial}{\partial y}(2xe^y)$

$= \cos x\frac{\partial}{\partial y}y + 2x\frac{\partial}{\partial y}e^y = \cos x.1 + 2xe^y = \cos x + 2xe^y$

and $\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}\left(\sin x + x^2e^y - 1\right) = \frac{\partial}{\partial x}\sin x + \frac{\partial}{\partial x}x^2e^y - \frac{\partial}{\partial x}1$

$= \cos x + e^y\frac{\partial}{\partial x}x^2 - 0 = \cos x + e^y(2x) = \cos x + 2xe^y$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Longrightarrow$ the given differential equation is exact.

Question-04 (Descriptive, Marks-05, Expected completion time= 10 mins.)

Determine whether the following initial value problem is exact? Also give its particular solution;

$(\sin 2x - \tan y)\,dx = x\sec^2 y dy; \qquad y(\pi) = \frac{\pi}{4}$.

Solution:

$(\sin 2x - \tan y)\,dx = x\sec^2 y dy \Longrightarrow (\sin 2x - \tan y)\,dx - x\sec^2 y dy = 0$

Comparing with Standard exact differential equation;

$M(x,y)dx + N(x,y)dy = 0.$
$\Longrightarrow M(x,y) = \sin 2x - \tan y$ and $N(x,y) = -x\sec^2 y$
$\Longrightarrow \frac{\partial M}{\partial y} = \frac{\partial}{\partial y}\left(\sin 2x - \tan y\right)$
$= \frac{\partial}{\partial y}\sin 2x - \frac{\partial}{\partial y}\tan y = 0 - \sec^2 y = \sec^2 y$
and $\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}\left(-x\sec^2 y\right) = \sec^2 y \frac{\partial}{\partial x}(-x)$
$= \sec^2 y$
$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Longrightarrow$ the given differential equation is exact and its solution is given by;

$$\underbrace{\int M(x,y)dx}_{y-\text{constant}} + \int \left(\text{terms of } N \text{ not containing } x\right) dy = C$$

$$\Longrightarrow \underbrace{\int \left(\sin 2x - \tan y\right) dx}_{y-\text{constant}} + \int (0)\, dy = C$$

$\Longrightarrow \int \sin 2x dx - \int \tan y dx + k = C$
$\Longrightarrow \frac{1}{2}\int 2\sin 2x dx - \tan y \int dx = C - k$
$\Longrightarrow \frac{1}{2}\left(-\cos 2x\right) - x\tan y = K - - - (1)$, where $C - k = K$ say!
Put $y(\pi) = \frac{\pi}{4}$ above;
$\frac{1}{2}\left(-\cos 2\pi\right) - \pi \tan\frac{\pi}{4} = K$
$\Longrightarrow \frac{1}{2}(-1) - \pi 1 = K \Longrightarrow -\pi - \frac{1}{2} = K - - -$ Put in (1)
$\therefore \frac{1}{2}\left(-\cos 2x\right) - x\tan y = -\pi - \frac{1}{2}$
$\Longrightarrow \cos 2x + 2x\tan y = 2\pi + 1$
$\Longrightarrow 2x\tan y = -\cos 2x + 2\pi + 1$
$\Longrightarrow \tan y = \frac{1}{2x}\left(1 + 2\pi - 2\cos 2x\right)$
$\Longrightarrow \boxed{y = \tan^{-1}\left[\frac{1}{2x}\left(1 + 2\pi - 2\cos 2x\right)\right]}$, is the required particular solution.

Lecture 6

Which of the following can be the integrating factor for the non-exact differential equation: $ydx - xdy = 0$?

$\frac{1}{y^2}$

$\frac{1}{x^2}$

$\frac{1}{xy}$

$\frac{1}{x^2-y^2}$

$\frac{1}{x^2+y^2}$

All of these (*Correct*)

Solution

i) $ydx - xdy = 0$

Multiplying with $\frac{1}{y^2} \Longrightarrow \frac{y}{y^2}dx - \frac{x}{y^2}dy = 0 \Longrightarrow \frac{1}{y}dx - \frac{x}{y^2}dy = 0$

Comparing with the exact form of differential equation:

$M(x,y)dx + N(x,y)dy = 0$

$\Longrightarrow M(x,y) = \frac{1}{y}$ and $N(x,y) = -\frac{x}{y^2}$

$\Longrightarrow \frac{\partial}{\partial y}M(x,y) = \frac{\partial}{\partial y}\left(\frac{1}{y}\right) = -\frac{1}{y^2}$

and $\frac{\partial}{\partial x}N(x,y) = \frac{\partial}{\partial x}\left(-\frac{x}{y^2}\right) = -\frac{1}{y^2}\frac{\partial}{\partial x}(x) = -\frac{1}{y^2}$

$\therefore \frac{\partial}{\partial y}M(x,y) = -\frac{1}{y^2} = \frac{\partial}{\partial x}N(x,y)$

$\Longrightarrow \frac{1}{y^2}$ is an integrating factor of $ydx - xdy = 0$.

ii) $ydx - xdy = 0$

Multiplying with $\frac{1}{x^2} \Longrightarrow \frac{y}{x^2}dx - \frac{x}{x^2}dy = 0 \Longrightarrow \frac{y}{x^2}dx - \frac{1}{x}dy = 0$

Comparing with the exact form of differential equation:

$M(x,y)dx + N(x,y)dy = 0$

$\Longrightarrow M(x,y) = \frac{y}{x^2}$ and $N(x,y) = -\frac{1}{x}$

$\Longrightarrow \frac{\partial}{\partial y}M(x,y) = \frac{\partial}{\partial y}\left(\frac{y}{x^2}\right) = \frac{1}{x^2}\frac{\partial}{\partial y}(y) = \frac{1}{x^2}$

and $\frac{\partial}{\partial x}N(x,y) = \frac{\partial}{\partial x}\left(-\frac{1}{x}\right) = \frac{1}{x^2}$

$\therefore \frac{\partial}{\partial y}M(x,y) = \frac{1}{x^2} = \frac{\partial}{\partial x}N(x,y)$

$\Longrightarrow \frac{1}{x^2}$ is an integrating factor of $ydx - xdy = 0$.

iii) $ydx - xdy = 0$

Multiplying with $\frac{1}{xy} \Longrightarrow \frac{y}{xy}dx - \frac{x}{xy}dy = 0 \Longrightarrow \frac{1}{x}dx - \frac{1}{y}dy = 0$

Comparing with the exact form of differential equation:

$M(x,y)dx + N(x,y)dy = 0$

$\Longrightarrow M(x,y) = \frac{1}{x}$ and $N(x,y) = -\frac{1}{y}$

$\Longrightarrow \frac{\partial}{\partial y}M(x,y) = \frac{\partial}{\partial y}\left(\frac{1}{x}\right) = 0$ and $\frac{\partial}{\partial x}N(x,y) = \frac{\partial}{\partial x}\left(-\frac{1}{y}\right) = 0$

$\therefore \frac{\partial}{\partial y}M(x,y) = 0 = \frac{\partial}{\partial x}N(x,y)$

$\Longrightarrow \frac{1}{xy}$ is an integrating factor of $ydx - xdy = 0$.

iv) $ydx - xdy = 0$

Multiplying with $\frac{1}{x^2-y^2} \Longrightarrow \frac{y}{x^2-y^2}dx - \frac{x}{x^2-y^2}dy = 0$

Comparing with the exact form of differential equation:

$M(x,y)dx + N(x,y)dy = 0$

$\Longrightarrow M(x,y) = \frac{y}{x^2-y^2}$ and $N(x,y) = -\frac{x}{x^2-y^2}$

$\Longrightarrow \frac{\partial}{\partial y}M(x,y) = \frac{\partial}{\partial y}\left(\frac{y}{x^2-y^2}\right) = \frac{\left(x^2-y^2\right)\frac{\partial}{\partial y}y - y\frac{\partial}{\partial y}\left(x^2-y^2\right)}{(x^2-y^2)^2}$

$= \frac{\left(x^2-y^2\right)1-y(-2y)}{(x^2-y^2)^2} = \frac{x^2-y^2+2y^2}{(x^2-y^2)^2} = \frac{x^2+y^2}{(x^2-y^2)^2}$

and $\frac{\partial}{\partial x}N(x,y) = \frac{\partial}{\partial x}\left(-\frac{x}{x^2-y^2}\right) = -\frac{\left(x^2-y^2\right)\frac{\partial}{\partial x}x - x\frac{\partial}{\partial x}\left(x^2-y^2\right)}{(x^2-y^2)^2}$

$= -\frac{\left(x^2-y^2\right)1-x(2x)}{(x^2-y^2)^2} = -\frac{x^2-y^2-2x^2}{(x^2-y^2)^2} = -\frac{\left(-x^2-y^2\right)}{(x^2-y^2)^2} = \frac{x^2+y^2}{(x^2-y^2)^2}$

$\therefore \frac{\partial}{\partial y}M(x,y) = \frac{x^2+y^2}{(x^2-y^2)^2} = \frac{\partial}{\partial x}N(x,y)$

$\Longrightarrow \frac{1}{x^2-y^2}$ is an integrating factor of $ydx - xdy = 0$.

v) $ydx - xdy = 0$

Multiplying with $\frac{1}{x^2+y^2} \Longrightarrow \frac{y}{x^2+y^2}dx - \frac{x}{x^2+y^2}dy = 0$

Comparing with the exact form of differential equation:

$M(x,y)dx + N(x,y)dy = 0$

$\Longrightarrow M(x,y) = \frac{y}{x^2+y^2}$ and $N(x,y) = -\frac{x}{x^2+y^2}$

$\Longrightarrow \frac{\partial}{\partial y}M(x,y) = \frac{\partial}{\partial y}\left(\frac{y}{x^2+y^2}\right) = \frac{\left(x^2+y^2\right)\frac{\partial}{\partial y}y - y\frac{\partial}{\partial y}\left(x^2+y^2\right)}{(x^2+y^2)^2}$

$= \frac{\left(x^2+y^2\right)1-y(2y)}{(x^2+y^2)^2} = \frac{x^2-y^2}{(x^2+y^2)^2}$

and $\frac{\partial}{\partial x}N(x,y) = \frac{\partial}{\partial x}\left(-\frac{x}{x^2+y^2}\right) = -\frac{\left(x^2+y^2\right)\frac{\partial}{\partial x}x - x\frac{\partial}{\partial x}\left(x^2+y^2\right)}{(x^2+y^2)^2}$

$= -\frac{\left(x^2+y^2\right)1-x(2x)}{(x^2+y^2)^2} = -\frac{y^2-x^2}{(x^2+y^2)^2} = \frac{x^2-y^2}{(x^2+y^2)^2} = \frac{x^2-y^2}{(x^2+y^2)^2}$

$\therefore \frac{\partial}{\partial y}M(x,y) = \frac{x^2-y^2}{(x^2+y^2)^2} = \frac{\partial}{\partial x}N(x,y) \Longrightarrow \frac{1}{x^2+y^2}$ is an integrating factor of $ydx - xdy = 0$.

Question-02 (Descriptive, Marks-02,Expected completion time=4 mins)

Determine whether e^x is an integrating factor for the non-exact differential equation:$y(x+y+1)dx + (x+2y)dy = 0$?

Solution

Multiplying e^x with given DE;$\Longrightarrow e^x y(x+y+1)dx + e^x(x+2y)dy = 0$

And comparing with exact form of DE: $M(x,y)dx + N(x,y)dy = 0$

$\Longrightarrow M(x,y) = e^x y(x+y+1)$ and $N(x,y) = e^x(x+2y)$

$\Longrightarrow \frac{\partial}{\partial y}M(x,y) = \frac{\partial}{\partial y}\left[e^x y(x+y+1)\right] = \frac{\partial}{\partial y}\left[y^2e^x + ye^x + xye^x\right]$

$= e^x\frac{\partial}{\partial y}\left[y^2+y+xy\right] = e^x\frac{\partial}{\partial y}\left[y^2+y+xy\right]$

$= e^x\left[\frac{\partial}{\partial y}y^2 + \frac{\partial}{\partial y}y + \frac{\partial}{\partial y}xy\right] = e^x\left(2y+1+x\frac{\partial}{\partial y}y\right)$

$= e^x\left(2y+1+x\right)$

And $\frac{\partial}{\partial x}N(x,y) = \frac{\partial}{\partial x}\left[e^x(x+2y)\right] = e^x\frac{\partial}{\partial x}(x+2y) + (x+2y)\frac{\partial}{\partial x}e^x$

$= e^x\left(1\right) + (x+2y)e^x = e^x\left(x+2y+1\right)$

$\therefore \frac{\partial}{\partial y}M(x,y) = e^x\left(x+2y+1\right) = \frac{\partial}{\partial x}N(x,y) \Longrightarrow e^x$ is an integrating factor for non-exact differential equation.

Question-03 (Descriptive, Marks-03,Expected completion time = 6 mins)

Show that $\frac{1}{xM(x,y)-yN(x,y)}$ is an integrating factor for the non-exact differential equation:

$y\left(xy\sin xy+\cos xy\right)dx+x\left(xy\sin xy-\cos xy\right)dy=0$

Solution:

Here $M(x,y)=y\left(xy\sin xy+\cos xy\right)$ and $N(x,y)=x\left(xy\sin xy-\cos xy\right)$

$xM(x,y)-yN(x,y)=x\left[y\left(xy\sin xy+\cos xy\right)\right]-y\left[x\left(xy\sin xy-\cos xy\right)\right]$

$=x\left[xy^2\sin xy+y\cos xy\right]-y\left[x^2y\sin xy-x\cos xy\right]$

$=x^2y^2\sin xy+xy\cos xy-x^2y^2\sin xy+xy\cos xy$

$=2xy\cos xy$

$\therefore \frac{1}{xM(x,y)-yN(x,y)}=\frac{1}{2xy\cos xy}$ multiplying this with given DE.

$\Longrightarrow \frac{1}{2xy\cos xy}y\left(xy\sin xy+\cos xy\right)dx+\frac{1}{2xy\cos xy}x\left(xy\sin xy-\cos xy\right)dy=0$

$\Longrightarrow \frac{1}{2x\cos xy}\left(\cos xy+xy\sin xy\right)dx+\frac{1}{2y\cos xy}\left(xy\sin xy-\cos xy\right)dy=0$

$\Longrightarrow \frac{1}{2}\left(\frac{\cos xy}{x\cos xy}+\frac{xy\sin xy}{x\cos xy}\right)dx+\frac{1}{2}\left(\frac{xy\sin xy}{y\cos xy}-\frac{\cos xy}{y\cos xy}\right)dy=0$

$\Longrightarrow \left(\frac{1}{x}+y\tan xy\right)dx+\left(x\tan xy-\frac{1}{y}\right)dy=0$

Now $M(x,y)=\frac{1}{x}+y\tan xy$ and $N(x,y)=x\tan xy-\frac{1}{y}$

$\Longrightarrow \frac{\partial}{\partial y}M(x,y)=\frac{\partial}{\partial y}\left(\frac{1}{x}+y\tan xy\right)=\frac{\partial}{\partial y}\left(\frac{1}{x}\right)+\frac{\partial}{\partial y}\left(y\tan xy\right)$

$=0+\frac{\partial}{\partial y}\left(y\tan xy\right)=y\frac{\partial}{\partial y}\left(\tan xy\right)+\tan xy\frac{\partial}{\partial y}\left(y\right)$

$=y\sec^2 xy\frac{\partial}{\partial y}\left(xy\right)+\tan xy\left(1\right)$

$=\left(y\sec^2 xy\right)x\frac{\partial}{\partial y}\left(y\right)+\tan xy=\left(y\sec^2 xy\right)x\left(1\right)+\tan xy$

$=xy\sec^2 xy+\tan xy$

And $\frac{\partial}{\partial x}N(x,y)=\frac{\partial}{\partial x}\left(x\tan xy-\frac{1}{y}\right)=\frac{\partial}{\partial x}\left(x\tan xy\right)-\frac{\partial}{\partial x}\left(\frac{1}{y}\right)$

$=x\frac{\partial}{\partial x}\left(\tan xy\right)+\tan xy\frac{\partial}{\partial x}x-0$

$=x\left(\sec^2 xy\right)\frac{\partial}{\partial x}xy+\tan xy\left(1\right)$

$=x\left(\sec^2 xy\right)y\frac{\partial}{\partial x}x+\tan xy$

$=x\left(\sec^2 xy\right)y\left(1\right)+\tan xy$

$=xy\sec^2 xy+\tan xy$

$\therefore \frac{\partial}{\partial y}M(x,y)=xy\sec^2 xy+\tan xy=\frac{\partial}{\partial x}N(x,y)$

$\Longrightarrow \frac{1}{xM(x,y)-yN(x,y)}$ is an integrating factor for the non-exact differential equation.

Question-04 (Descriptive, Marks-05,Expected completion time = 10 mins)

Solve the following differential equation by verifying that $e^{\int \frac{1}{M}\left(\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}\right)dy}$ be its an integrating factor.

$ydx+\left(3+3x-y\right)dy=0.$

Solution:

Given that $M(x,y)=y$ and $N(x,y)=3+3x-y$

$\therefore \frac{1}{M}\left(\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}\right)=\frac{1}{y}\left(\frac{\partial}{\partial x}\left(3+3x-y\right)-\frac{\partial}{\partial y}y\right)$

$=\frac{1}{y}\left(3-1\right)=\frac{2}{y}$

$\Longrightarrow e^{\int \frac{1}{M}\left(\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}\right)dy}=e^{\int \frac{2}{y}dy}=e^{2\int \frac{1}{y}dy}$

$=e^{2\ln y}=e^{\ln y^2}=y^2$,multiplying this by the given DE.

$\therefore y^2\left[ydx+\left(3+3x-y\right)dy\right]=0$

$\Longrightarrow y^3dx+\left(3y^2+3xy^2-y^3\right)dy=0$

Now $M(x,y)=y^3$ and $N(x,y)=3y^2+3xy^2-y^3$

$\implies \frac{\partial}{\partial y}M(x,y) = \frac{\partial}{\partial y}y^3 = 3y^2$

and $\frac{\partial}{\partial x}N(x,y) = \frac{\partial}{\partial x}\left(3y^2 + 3xy^2 - y^3\right) = 3y^2$

$\therefore \frac{\partial}{\partial y}M(x,y) = 3y^2 = \frac{\partial}{\partial x}N(x,y)$

$\implies e^{\int \frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)dy}$ is an integrating factor of given DE, whose solution is given by;

$$\underbrace{\int M(x,y)dx}_{y-\text{constant}} + \int \left(\text{terms of } N \text{ not containing } x\right)dy = C$$

$$\implies \underbrace{\int y^3dx}_{y-\text{constant}} + \int \left(3y^2 - y^3\right)dy = C$$

$\implies y^3\int dx + \int 3y^2dy - \int y^3dy = C$

$\implies y^3x + 3\int y^2dy - \frac{1}{4}y^4 = C$

$\implies y^3x + 3\left(\frac{1}{3}y^3\right) - \frac{1}{4}y^4 = C$

$\therefore \boxed{y^3x + y^3 - \frac{1}{4}y^4 = C}$, is the required solution.

1 (*MCQ, marks-1, expected completion time 1min*)
The integrating factor for the following 1st order linear equation is———.
$\left(1+y^2\right)dx = \left(\tan^{-1}y - x\right)dy$
$e^{\tan^{-1}x}$
$e^{\tan^{-1}y}$ (*correct*)
$e^{\frac{1}{1+x^2}}$
$e^{\frac{1}{1+y^2}}$

Solution:
$\left(1+y^2\right)dx = \left(\tan^{-1}y - x\right)dy$
$\because y^2$ appears in the equation, so we make the DE linear in x,
$\therefore \frac{dx}{dy} = \frac{\left(\tan^{-1}y - x\right)}{(1+y^2)}$
$\implies \frac{dx}{dy} = \frac{\tan^{-1}y}{1+y^2} - \frac{x}{1+y^2}$
$\implies \frac{dx}{dy} + \frac{1}{1+y^2}x = \frac{\tan^{-1}y}{1+y^2}$
Comparing with standard linear differential equation in x i.e.
$\frac{dx}{dy} + P(y)x = Q(y)$
$\implies$ the integrating factor:
$I.F = e^{\int P(y)dy} = e^{\int \frac{1}{1+y^2}dy}$
$= e^{\tan^{-1}y} \qquad \left\{\because \int \frac{f'(x)}{a^2+[f(x)]^2}dx = \frac{1}{a}\tan^{-1}\left(\frac{f(x)}{a}\right)\right.$

2) (*MCQ, marks-1, expected completion time 1min*)
For the 1st order linear differential equation: $x\left(\frac{dy}{dx}+y\right)+y=1$, the continuos function $q(x) = - - --$, while comparing this with standard form $\frac{dy}{dx}+p(x)y = q(x)$
$1+\frac{1}{x}$
$\frac{1}{x}$(*correct*)
$1-\frac{1}{x}$
$-\frac{1}{x}$
Solution:
$x\left(\frac{dy}{dx}+y\right)+y=1$
$\implies \left(\frac{dy}{dx}+y\right)+\frac{1}{x}y = \frac{1}{x} \implies \frac{dy}{dx}+\left(x+\frac{1}{x}\right)y = \frac{1}{x}$
comparing this with standard form $\frac{dy}{dx}+p(x)y = q(x)$
$\implies p(x) = x+\frac{1}{x}$ and $q(x) = \frac{1}{x}$

(*Descriptive, marks-2, expected completion time 4min*)
3) Find the integrating factor of 1st order linear differential equation:
$e^x\left[y-3\left(e^x+1\right)^2\right]dx + \left(e^x+1\right)dy = 0.$
Solution:
$e^x\left[y-3\left(e^x+1\right)^2\right]dx + \left(e^x+1\right)dy = 0$
$\implies \left(e^x+1\right)dy = -e^x\left[y-3\left(e^x+1\right)^2\right]dx$
$\implies \frac{dy}{dx} = -\frac{e^x}{(e^x+1)}\left[y-3\left(e^x+1\right)^2\right]$

$\Longrightarrow \frac{dy}{dx} = -\frac{e^x}{(e^x+1)}y + 3\frac{e^x}{(e^x+1)}\left(e^x+1\right)^2$
$\Longrightarrow \frac{dy}{dx} + \frac{e^x}{(e^x+1)}y = 3e^x\left(e^x+1\right)$
comparing this with standard form $\frac{dy}{dx} + p(x)y = q(x)$
$\Longrightarrow p(x) = \frac{e^x}{(e^x+1)}$
$\therefore$ the integrating factor:
$IF = e^{\int p(x)dx} = e^{\int \frac{e^x}{(e^x+1)}dx}$
$\left\{\because \int \frac{f'(x)}{f(x)}dx = \ln|f(x)|\right.$, where $f(x) = e^x + 1$ and $f'(x) = e^x$

$= e^{\ln|e^x+1|} = e^x + 1$

(*Descriptive, marks-3, expected completion time 6min)*
4) Convert the following differential equation into standard linear form:
$\left(x + 2y^3\right)\frac{dy}{dx} = y.$
Solution:
$\because y^3$ appears in the equation, so we make the DE linear in x
$\left(x + 2y^3\right)\frac{dy}{dx} = y$
$\Longrightarrow \frac{dy}{dx} = \frac{y}{x+2y^3}$
$\Longrightarrow \frac{dx}{dy} = \frac{x+2y^3}{y}$
$\Longrightarrow \frac{dx}{dy} = \frac{x}{y} + \frac{2y^3}{y}$
$\Longrightarrow \frac{dx}{dy} + \left(\frac{-1}{y}\right)x = 2y^2$
Comparing with standard linear differential equation in x i.e.
$\frac{dx}{dy} + P(y)x = Q(y)$
$\Longrightarrow P(y) = \frac{-1}{y}$ and $Q(y) = 2y^2$.

5 (*Descriptive, marks-5, expected completion time 10min)*
Solve $\left(1 + x^2\right)\frac{dy}{dx} + y = \tan^{-1}x$
Solution:
$\left(1 + x^2\right)\frac{dy}{dx} + y = \tan^{-1}x$
$\Longrightarrow \frac{dy}{dx} + \left(\frac{1}{1+x^2}\right)y = \frac{\tan^{-1}x}{1+x^2}$
comparing this with standard form $\frac{dy}{dx} + p(x)y = q(x)$
$\Longrightarrow p(x) = \frac{1}{1+x^2}$ and $q(x) = \frac{\tan^{-1}x}{1+x^2}$
$\therefore$Integrating factor:
$= IF = e^{\int p(x)dx} = e^{\int \frac{1}{1+x^2}dx}$

$$= e^{\tan^{-1}x} \qquad \left\{\begin{array}{l} \because \int \frac{f'(x)}{a^2+[f(x)]^2}dx = \frac{1}{a}\tan^{-1}\left(\frac{f(x)}{a}\right) \\ \text{here } f(x) = x \Longrightarrow f'(x) = 1 \text{ and } a = 1 \end{array}\right.$$

$\therefore$its solution is given by;
(Integrating Factor)$\times$(Dependent variable) $= \int$ ((Integrating Factor) $\times$ RHS) dx
$\Longrightarrow e^{\tan^{-1}x}y = \int e^{\tan^{-1}x}\frac{\tan^{-1}x}{1+x^2}dx + C - - - (1)$
Put $u = \tan^{-1}x \Longrightarrow \frac{du}{dx} = \frac{1}{1+x^2}$
$\Longrightarrow du = \frac{1}{1+x^2}dx$

$\therefore \int e^{\tan^{-1}x}\frac{\tan^{-1}x}{1+x^2}dx = \int \tan^{-1}x e^{\tan^{-1}x}\frac{1}{1+x^2}dx = \int ue^u du$

Now integrating by parts taking u as 1st function and e^u as 2nd one and apply the following

$\int I.II du = I\int II du - \int \left(\left(\int II du\right)\frac{d}{du}I\right)du$

$\therefore \int ue^u du = u\int e^u du - \int \left(\left(\int e^u du\right)\frac{d}{du}u\right)du = u\left(e^u\right) - \int \left(\left(e^u\right)1\right)du$

$= ue^u - \int e^u du = ue^u - e^u = e^u\left(u-1\right) = e^{\tan^{-1}x}\left(\tan^{-1}x - 1\right)$

$\therefore (1) \implies e^{\tan^{-1}x}y = e^{\tan^{-1}x}\left(\tan^{-1}x - 1\right) + C$

$\implies y = \frac{e^{\tan^{-1}x}}{e^{\tan^{-1}x}}\left(\tan^{-1}x - 1\right) + \frac{C}{e^{\tan^{-1}x}}$

$\implies \boxed{y = \left(\tan^{-1}x - 1\right) + Ce^{-\tan^{-1}x}}$, $\quad \left\{\because \frac{1}{e^t} = e^{-t}\right.$

which is the required solution, where C is the constant of integration.

1) (*MCQ, marks-01, expected completion time 01min.*)
In the Bernoulli equation: $x\frac{dy}{dx} + y = y^2 \log x$, the $p(x), q(x)$ and n are —— —— respectively.
$x, 1$ and 2
$\frac{1}{x}, \log x$ and 2
$\frac{1}{x}, \frac{\log x}{x}$ and 2 (correct)
$x, \log x$ and 2

Solution:
Given that $x\frac{dy}{dx} + y = y^2 \log x$
$\Longrightarrow \frac{dy}{dx} + \frac{1}{x}y = \left(\frac{\log x}{x}\right) y^2$
Comparing this with the general form of Bernoulli equation:
$\frac{dy}{dx} + p(x)y = q(x)y^n$
$\Longrightarrow p(x) = \frac{1}{x}, q(x) = \frac{\log x}{x}$ and $n = 2$.

2) (*Descriptive, marks-02, expected completion time 04min.*)
Using a suitable substitution, reduce the Bernoulli differential equation:
$\sec^2 y\frac{dy}{dx} + x\tan y = x^3$ into linear 1st order differential equation.

Solution:
$\sec^2 y\frac{dy}{dx} + x\tan y = x^3$
Put $v = \tan y \Longrightarrow \frac{d}{dx}v = \frac{d}{dx}\tan y$
$\Longrightarrow \frac{dv}{dx} = \sec^2 y\frac{dy}{dx}$
$\therefore$the given differential equation becomes;
$\frac{dv}{dx} + xv = x^3$
Comparing this with standard form 1st order linear differential equation in dependent variable v;
$\frac{dv}{dx} + p(x)v = q(x)$, where $p(x) = x$ and $q(x) = x^3$.

3) (*Descriptive, marks-03, expected completion time 06min.*)
Using a suitable substitution, reduce the Bernoulli differential equation:
$x\frac{dy}{dx} + y = y^2 \log x$ into linear 1st order differential equation.

Solution:
$\because$the general form of Bernoulli equation is: $\frac{dy}{dx} + p(x)y = q(x)y^n$
For the given DE: $x\frac{dy}{dx} + y = y^2 \log x$,
dividing by xy^2 on both sides;
$\Longrightarrow \frac{1}{xy^2}\left(x\frac{dy}{dx} + y\right) = \frac{1}{xy^2}\left(y^2 \log x\right)$
$\Longrightarrow \frac{1}{y^2}\frac{dy}{dx} + \frac{1}{xy} = \frac{1}{x}\log x$
$\Longrightarrow y^{-2}\frac{dy}{dx} + \frac{1}{x}y^{-1} = \frac{1}{x}\log x - - - -(1)$
Put $v = y^{-1}$

$\Longrightarrow \frac{d}{dx}v = \frac{d}{dx}x^{-1} = -\frac{1}{y^2}\frac{d}{dx}y$

$\Longrightarrow \frac{1}{y^2}\frac{dy}{dx} = -\frac{dv}{dx}$

$\therefore (1) \Longrightarrow -\frac{dv}{dx} + \frac{1}{x}v = \frac{1}{x}\log x$

$\Longrightarrow \frac{dv}{dx} + \left(-\frac{1}{x}\right)v = -\frac{1}{x}\log x$

Comparing this with standard form of 1st order linear differential equation in dependent variable v;

$\frac{dv}{dx} + p(x)v = q(x)$, where $p(x) = -\frac{1}{x}$ and $q(x) = -\frac{1}{x}\log x$.

4) (*Descriptive, marks-05, expected completion time 10 min.*)

Solve the differential equation: $\frac{dy}{dx} + x\sin 2y = x^3\cos^2 y$.

Solution:

$\frac{dy}{dx} + x\sin 2y = x^3\cos^2 y$

Dividing both sides by $\cos^2 y$;

$\Longrightarrow \frac{1}{\cos^2 y}\left(\frac{dy}{dx} + x\sin 2y\right) = \frac{1}{\cos^2 y}\left(x^3\cos^2 y\right)$

$\Longrightarrow \frac{1}{\cos^2 y}\frac{dy}{dx} + \frac{1}{\cos^2 y}x\sin 2y = x^3$

$\Longrightarrow \sec^2 y\frac{dy}{dx} + \frac{1}{\cos^2 y}x\left(2\sin y\cos y\right) = x^3 \quad \because \sin 2\alpha = 2\sin\alpha\cos\alpha$

$\Longrightarrow \sec^2 y\frac{dy}{dx} + 2x\frac{\sin y}{\cos y} = x^3$

$\Longrightarrow \sec^2 y\frac{dy}{dx} + 2x\tan y = x^3 - - - -(1)$

Put $v = \tan y \Longrightarrow \frac{d}{dx}v = \frac{d}{dx}\tan y \Longrightarrow \frac{dv}{dx} = \sec^2 y\frac{dy}{dx}$

$\therefore (1) \Longrightarrow \frac{dv}{dx} + (2x)\,v = x^3$

Comparing this with standard form of 1st order linear differential equation in dependent variable v;

$\frac{dv}{dx} + p(x)v = q(x)$, where $p(x) = 2x$ and $q(x) = x^3$.

$\therefore$the integrating factor:

$IF = e^{\int p(x)dx} = e^{\int 2xdx} = e^{2\int xdx} = e^{2\left(\frac{1}{2}x^2\right)} = e^{x^2}$

$\therefore$its solution is given by;

(Integrating Factor)$\times$(Dependent variable) $= \int$ ((Integrating Factor) $\times$ RHS) dx

$\Longrightarrow e^{x^2}v = \int e^{x^2}x^3dx - - - -(2)$

$$\left\{\begin{array}{c}
\int e^{x^2}x^3dx = \int e^{x^2}x^2xdx \\
\text{put } t = x^2 \\
\Longrightarrow \text{on differentiating w.r.t } x \quad \frac{dt}{dx} = 2x \\
\Longrightarrow \frac{1}{2}dt = xdx \\
\therefore \int e^{x^2}x^2\left(xdx\right) = \int e^t t\left(\frac{1}{2}dt\right) = \frac{1}{2}\int e^t tdt \\
\text{integrating by parts by taking } t \text{ as 1st function and } e^t \text{ as 2nd and using:} \\
\int I.IIdx = I\left(\int IIdx\right) - \int\left(\left(\int IIdx\right)\frac{d}{dt}I\right)dt \\
\Longrightarrow \int te^tdt = t\int e^tdt - \int\left(\left(\int e^tdt\right)\left(\frac{d}{dt}t\right)\right)dt \\
\Longrightarrow \int te^tdt = t\left(e^t\right) - \int\left(\left(e^t\right)(1)\right)dt \\
\Longrightarrow \int te^tdt = te^t - \int e^tdt = te^t - e^t = e^t\left(t-1\right) = e^{x^2}\left(x^2-1\right) \\
\therefore \int e^{x^2}x^2\left(xdx\right) = \frac{1}{2}e^{x^2}\left(x^2-1\right)
\end{array}\right.$$

Now (2)$\Longrightarrow e^{x^2}v = \frac{1}{2}e^{x^2}\left(x^2-1\right) + C$, where C is the constant of integration.

$\Longrightarrow e^{x^2}\tan y = \frac{1}{2}e^{x^2}\left(x^2-1\right) + C$

$\Longrightarrow \tan y = \frac{1}{e^{x^2}}\left(\frac{1}{2}e^{x^2}\left(x^2-1\right) + C\right)$

$$\implies \tan y = \frac{1}{2}\left(x^2 - 1\right) + Ce^{-x^2}$$

$$\implies \boxed{y = \tan^{-1}\left(\frac{1}{2}\left(x^2 - 1\right) + Ce^{-x^2}\right)}$$

1) MCQ

If $\frac{dy}{dx} = y$, then $y = ---$.

x

$\log x$

e^x

ke^x (*correct*)

Solution:

$\frac{dy}{dx} = y$

Separating the variables;

$\frac{dy}{y} = dx$

Integrating

$\int \frac{dy}{y} = \int dx$

$\implies \ln y = x + C$,where C is the constant of integration.

Taking Anti-log on both sides;

$e^{\ln y} = e^{x+C}$

$\implies y = e^x e^C \quad \because e^{\ln t} = t \quad \forall t \in \mathbb{R}_+$

$\implies y = ke^x$,where $k = e^C \because$ *(Constant)*^{Constant}=*Constant*

2) MCQ

For a Ist order linear differential equation: $\cos^3 x \frac{dy}{dx} + y \cos x = \sin x$, the integrating factor $\mu(x) = ---$.

$e^{\cot x}$

$e^{\tan x}$ (*correct*)

$e^{-\tan x}$

$e^{-\cot x}$

Solution:

$\cos^3 x \frac{dy}{dx} + y \cos x = \sin x$

$\implies \frac{dy}{dx} + \frac{1}{\cos^3 x} y \cos x = \frac{1}{\cos^3 x} \sin x$

$\implies \frac{dy}{dx} + \frac{1}{\cos^2 x} y = \frac{1}{\cos^3 x} \sin x$

Comparing with standard linear differential equation of 1st order;

$\frac{dy}{dx} + p(x)y = q(x)$, where $p(x) = \frac{1}{\cos^2 x} = \sec^2 x$

$\therefore$ integrating factor $= \mu(x) = e^{\int \sec^2 x dx} = e^{\tan x}$

3) Descriptive of marks 2

Solve the differential equation: $\ln\left(\frac{dy}{dx}\right) = y$

Solution:

$\ln\left(\frac{dy}{dx}\right) = y$

Taking Anti-log on both sides;

$\implies e^{\ln\left(\frac{dy}{dx}\right)} = e^y$

$\frac{dy}{dx} = e^y \quad \because e^{\ln t} = t \quad \forall t \in \mathbb{R}_+$

Separating the variables;
$e^{-y}dy = dx$
Integrating;
$\int e^{-y}dy = \int dx$
$\implies -e^{-y} = x + C$
$\implies e^{-y} = -x - C$
Taking ln;
$\ln e^{-y} = \ln(k - x)$, where $k = -C$ as *-constant=constant*
$\implies -y = \ln(k - x)$
$\implies \boxed{y = -\ln(k - x)}$

4) Descriptive of marks 3
Convert the differential equation: $\frac{dy}{dx}(x^2y^3 + xy) = 1$ into linear form.
Solution:
$\frac{dy}{dx}(x^2y^3 + xy) = 1$
$\implies x^2y^3 + xy = \frac{dx}{dy}$
$\implies \frac{dx}{dy} = x^2y^3 + xy$
$\implies \frac{dx}{dy} - yx = x^2y^3$
i.e. the Bernoulli's equation. So dividing by x^2,
$\therefore \frac{1}{x^2}\left(\frac{dx}{dy} - yx\right) = \frac{1}{x^2}\left(x^2y^3\right)$
$\implies \frac{1}{x^2}\frac{dx}{dy} - \frac{1}{x^2}yx = y^3$
$\implies x^{-2}\frac{dx}{dy} - x^{-1}y = y^3 - - - (1)$
Put $v = x^{-1} \implies$ on differentiating with respect to y;
$\frac{d}{dy}v = \frac{d}{dy}x^{-1} = -x^{-2}\frac{dx}{dy} \implies x^{-2}\frac{dx}{dy} = -\frac{dv}{dy}$
$\therefore (1) \implies -\frac{dv}{dy} - yv = y^3$
$\boxed{\frac{dv}{dy} + yv = -y^3}$
i.e. Linear 1st order differential of the form $\frac{dv}{dy} + p(y)v = q(y)$, where $p(y) = y$ and $q(y) = -y^3$.

Descriptive of marks 5
Solve $\frac{x+y}{y-1}dx - \frac{1}{2}\left(\frac{x+1}{y-1}\right)^2 dy = 0$
Solution:
$\frac{x+y}{y-1}dx - \frac{1}{2}\left(\frac{x+1}{y-1}\right)^2 dy = 0$
Comparing with standard form of homogenous differential equation:
$M(x,y)dx + N(x,y)dy = 0$
$\implies M(x,y) = \frac{x+y}{y-1}$ and $N(x,y) = -\frac{1}{2}\left(\frac{x+1}{y-1}\right)^2$
$\implies \frac{\partial M}{\partial y} = \frac{\partial}{\partial y}\left(\frac{x+y}{y-1}\right) = \frac{(y-1)\frac{\partial}{\partial y}(x+y) - (x+y)\frac{\partial}{\partial y}(y-1)}{(y-1)^2}$
$= \frac{(y-1)\left(\frac{\partial}{\partial y}x + \frac{\partial}{\partial y}y\right) - (x+y)\left(\frac{\partial}{\partial y}y - \frac{\partial}{\partial y}1\right)}{(y-1)^2} = \frac{y-1-x-y}{(y-1)^2} = \frac{-(x+1)}{(y-1)^2}$ And;
$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}\left(-\frac{1}{2}\left(\frac{x+1}{y-1}\right)^2\right) = -\frac{1}{2}\frac{\partial}{\partial x}\left(\frac{x+1}{y-1}\right)^2 = -\frac{1}{2}2\left(\frac{x+1}{y-1}\right)^{2-1}\frac{\partial}{\partial x}\left(\frac{x+1}{y-1}\right)$

$\frac{\partial N}{\partial x} = -1\left(\frac{x+1}{y-1}\right)^1 \frac{\partial}{\partial x}\left(\frac{x+1}{y-1}\right) = -\left(\frac{x+1}{y-1}\right)\frac{\partial}{\partial x}\left(\frac{1}{y-1}(x+1)\right)$
$\frac{\partial N}{\partial x} = -\left(\frac{x+1}{y-1}\right)\left(\frac{1}{y-1}\right)\frac{\partial}{\partial x}(x+1) = -\left(\frac{x+1}{y-1}\right)\left(\frac{1}{y-1}\right)(1) = \frac{-(x+1)}{(y-1)^2}$
$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Longrightarrow$ the given differential equation is exact and its solution is given by;

Here $N(x,y) = -\frac{1}{2}\left(\frac{x+1}{y-1}\right)^2 = -\frac{1}{2(y-1)^2}(x+1)^2 = -\frac{1}{2(y-1)^2}\left(x^2+2x+1\right)$
$N(x,y) = \left(-\frac{x^2}{2(y-1)^2} - \frac{2x}{2(y-1)^2} - \frac{1}{2(y-1)^2}\right)$
$\underbrace{\int M(x,y)dx}_{y-\text{constant}} + \int(\text{terms of } N \text{ not containing } x)\, dy = C$
$\Longrightarrow \underbrace{\int\left(\frac{x+y}{y-1}\right)dx}_{y-\text{constant}} + \int\left(-\frac{1}{2(y-1)^2}\right)dy = C$
$\Longrightarrow \int\left(\frac{x+y}{y-1}\right)dx - \frac{1}{2}\int(y-1)^{-2}\,dy = C$
$\Longrightarrow \left(\frac{1}{y-1}\right)\left(\int(x+y)\,dx\right) - \frac{1}{2}\left(\frac{(y-1)^{-2+1}}{-2+1}\right) = C$
$\Longrightarrow \left(\frac{1}{y-1}\right)\left(\int xdx + \int ydx\right) - \frac{1}{2}\left(-\frac{1}{y-1}\right) = C$
$\Longrightarrow \left(\frac{1}{y-1}\right)\left(\frac{1}{2}x^2 + \int dx\right) + \frac{1}{2(y-1)} = C$
$\Longrightarrow \left(\frac{1}{y-1}\right)\left(\frac{1}{2}x^2 + x\right) + \frac{1}{2(y-1)} = C$
$\Longrightarrow \left(\frac{1}{y-1}\right)\left(\frac{1}{2}x^2 + x + \frac{1}{2}\right) = C$
$\Longrightarrow \left(\frac{1}{y-1}\right)\left(\frac{x^2+2x+1}{2}\right) = C$
$\Longrightarrow x^2+2x+1 = 2C(y-1)$
$\Longrightarrow (x+1)^2 = 2C(y-1)$
$\Longrightarrow y-1 = \frac{(x+1)^2}{2C}$
$\Longrightarrow y = \frac{(x+1)^2}{2C} + 1$
$\Longrightarrow y = \frac{1}{2C}(x+1)^2 + 1$
$\Longrightarrow y = k(x+1)^2 + 1$, where $k = \frac{1}{2C}$ as $\frac{1}{2constant}$ =*constant.*

1) MCQ, Marks-1, time 1min
The orthogonal trajectory to the straight line $ax + by = k$ is ——-.
$ax - by = k$
$by - ax = k$
$ay + bx = k$

$ay - bx = k$ (*correct*)
Solution:
$ax + by = k$
$\Longrightarrow by = k - ax$
$\Longrightarrow y = \frac{k-ax}{b} = \frac{k}{b} - \frac{ax}{b} = \frac{k}{b} - \frac{a}{b}x$
$\Longrightarrow \frac{dy}{dx} = -\frac{a}{b} = m =$slope of given line
$\Longrightarrow$Slope of orthogonal family $m_{\perp} = \frac{b}{a}$
$\therefore$corresponding DE of orthogonal trajectory;
$\frac{dy}{dx} = m_{\perp} = \frac{b}{a} \Longrightarrow ady = bdx$
$\Longrightarrow \int ady = \int bdx \Longrightarrow a\int dy = b\int dx$
$\Longrightarrow ay = bx + c$, where c is the constant of integration.
$\therefore \boxed{ay - bx = c}$, is the required orthogonal family.

2) Descriptive, Marks-2, time 4min
Find the slope of line orthogonal to the curve: $y = e^x \cosh x$ at point $(0, 1)$.
Solution:
$y = e^x \cosh x$
$\Longrightarrow y = e^x\left(\frac{e^x+e^{-x}}{2}\right) \qquad \because \cosh x = \frac{e^x+e^{-x}}{2}$
$\Longrightarrow y = \frac{e^xe^x+e^xe^{-x}}{2}$
$\Longrightarrow y = \frac{e^{2x}+1}{2} = \frac{e^{2x}+1}{2} = \frac{1}{2} + \frac{1}{2}e^{2x}$
$\Longrightarrow$Slope of given curve $= m = \frac{d}{dx}y = \frac{d}{dx}\left(\frac{1}{2} + \frac{1}{2}e^{2x}\right)$
$= \frac{d}{dx}\frac{1}{2} + \frac{d}{dx}\frac{1}{2}e^{2x} = 0 + \frac{1}{2}\frac{d}{dx}e^{2x} = \frac{1}{2}\left(2e^{2x}\right) = e^{2x}$
$\Longrightarrow$Slope of given curve at $(0, 1) = e^{2(0)} = e^0 = 1$
$\therefore$Slope of orthogonal curve at $(0, 1) = m_{\perp} = \frac{-1}{m} = -1$

3) Descriptive, Marks-3, time 6min
Find the differential equation satisfied by the family curves:$y = \sin x + k\cos x$, where is k an arbitrary constant.
Solution:
$y = \sin x + k\cos x - - - -(1)$
$\Longrightarrow \frac{d}{dx}y = \frac{d}{dx}(\sin x + k\cos x) = \frac{d}{dx}\sin x + \frac{d}{dx}k\cos x$
$\Longrightarrow y' = \cos x + k\frac{d}{dx}\cos x = \cos x + k(-\sin x) = \cos x - k\sin x - - - (2)$
Now eliminating k from (2), by substituting k in (2);
$(1) \Longrightarrow y - \sin x = k\cos x \Longrightarrow \frac{y-\sin x}{\cos x} = k$—put in (2)
$\therefore (2) \Longrightarrow y' = \cos x - \left(\frac{y-\sin x}{\cos x}\right)\sin x = \cos x - \left(\frac{y\sin x-\sin^2 x}{\cos x}\right)$

$\Longrightarrow y' = \frac{\cos^2 x - \left(y\sin x - \sin^2 x\right)}{\cos x} = \frac{\cos^2 x - y\sin x + \sin^2 x}{\cos x} = \frac{\left(\cos^2 x + \sin^2 x\right) - y\sin x}{\cos x}$

$\Longrightarrow y' = \frac{1 - y\sin x}{\cos x} \Longrightarrow \cos x y' = 1 - y\sin x$

$\therefore$required differential equation: $\boxed{\cos x \frac{dy}{dx} + y\sin x = 1}$

4) Descriptive, Marks-3, time 6min

If the population $p(t)$ of any community is growing at the rate given by $p(t) = p_0 e^{0.0534t}$, then how would long it take to four times?Here p_0 is the initial population.

Solution:

$\because p(t) = p_0 e^{0.0534t} - - - -(1)$

Let at time t,the population becomes 4 times, then $p(t) = 4p_0$

$\therefore (1) \Longrightarrow 4p_0 = p_0 e^{0.0534t}$

$\Longrightarrow 4 = e^{0.0534t}$

Taking ln on both sides;

$\ln 4 = \ln e^{0.0534t}$

$\Longrightarrow 1.386\,3 = (0.0534t)\ln e \qquad \because \ln x^y = y\ln x$

$\Longrightarrow 1.386\,3 = (0.0534t)\,1 \qquad\qquad \because \ln e = \log_e e = 1$

$\Longrightarrow t = \frac{1.386\,3}{0.0534} = 25.961 \approx 26$ years·

5) Descriptive, Marks-5, time 10min

If the population $p(t)$ of any community is growing at the rate given by $p(t) = p_0 e^{0.0534t}$, and after 4 years, its population is 1 million, then what would be its initial population p_0?

Solution:

$p(t) = p_0 e^{0.0534t} - - - -(1)$

Since after 4 years, population is 1 million.

$\therefore$put $t = 4$ and $p(t) = p(4) = 10^6$ in (1);

$\therefore (1) \Longrightarrow 10^6 = p_0 e^{0.0534(4)}$

$\Longrightarrow 10^6 = p_0 e^{0.213\,6}$

$\Longrightarrow p_0 = \frac{10^6}{e^{0.213\,6}} = 8.076\,7 \times 10^5$

6) Descriptive, Marks-5, time 10min

Find the orthogonal trajectory to the family of curves: $ay^2 = x^3$.

Solution:

Given curve family of curves: $ay^2 = x^3 - - - -(1)$

Differentiating w.r.t x

$\frac{d}{dx}ay^2 = \frac{d}{dx}x^3 \Longrightarrow a\frac{d}{dx}y^2 = 3x^2$

$\Longrightarrow a\left(2y\frac{d}{dx}y\right) = 3x^2 \Longrightarrow \frac{d}{dx}y = \frac{3x^2}{2ay} - - - -(2)$

From (1),put the value of a in (2)

$(1) \Longrightarrow a = \frac{x^3}{y^2} - - -$put in (2)

$\therefore (2) \Longrightarrow \frac{d}{dx}y = \frac{3x^2}{2\left(\frac{x^3}{y^2}\right)y} = \frac{3y}{2x}$

$\Longrightarrow$the differential equation associated with orthogonal family of the given curves:

$\frac{dy}{dx} = -\frac{2x}{3y}$

Now separating the variables;

$ydy = \frac{-2}{3} xdx$

Integrating;

$\Longrightarrow \int ydy = \int \frac{-2}{3} xdx = \frac{-2}{3} \int xdx$

$\Longrightarrow \frac{1}{2} y^2 = \frac{-2}{3} \left(\frac{1}{2} x^2 \right) + c$

$y^2 = \frac{2(-2)}{3} \left(\frac{1}{2} x^2 \right) + 2c$

$\Longrightarrow \boxed{y^2 = \frac{-2}{3} \left(x^2 \right) + k}$ i.e. the required family of curves, where $2c = k$.

Question-1 (MCQ, Marks-1, Lecture-13)
If the Wronskian of a set of functions is zero, then the set will be——.
Linearly independent *(correct)*
Linearly dependent
Neither dependent nor independent
None of these

Question-2 (MCQ, Marks-1, Lecture-14)
$W(x, x+1) =$———.
1
0
-1(*correct*)
$-x$

Question-3 (MCQ, Marks-1, Lecture-15)
If $y_1 = e^{-x}$ is the 1st solution of the differential equation:$y'' + 2y' + y = 0$, then which of the following will give second?
$xe^{-x} \int \frac{e^{-\int 2dx}}{e^{-2x}} dx$
$e^{-x} \int \frac{e^{-\int 2dx}}{e^{-2x}} dx \quad (correct)$
$e^{-x} \int \frac{e^{-2x}}{e^{-\int 2dx}} dx$
$xe^{-x} \int \frac{e^{-2x}}{e^{-\int 2dx}} dx$

Question-4 (Marks-2, Lecture-13)
Show that the functions $f(x) = \sin x$ and $g(x) = 2\sin x$ are linearly dependent.
Solution:
$W(f, g) = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = \begin{vmatrix} \sin x & 2\sin x \\ \cos x & 2\cos x \end{vmatrix} = 2\sin x\cos x - 2\sin x\cos x =$ $0 \Longrightarrow$ the given functions are linearly dependent.

Question-5 (Marks-2, Lecture-14)
Show that solutions $f(x) = e^x$ and $g(x) = e^{-x}$ of the differential equation: $y'' - y = 0$, will form the fundamental set.
Solution:
$$\because W(f, g) = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -e^x e^{-x} - e^x e^{-x} = -2 \neq 0$$
$\Longrightarrow \{f, g\}$ is linearly independent and hence will form a fundamental set.

Question-6 (Marks-2, Lecture-15)
Evaluate $e^{-\int p(x)dx}$ by using the $\cos x.y'' - \sin x.y' + e^x y = 0$ by writing it in the standard form: $y'' + p(x)y' + q(x).y = 0$.
Solution:
$\cos x.y'' - \sin x.y' + e^x y = 0 \Longrightarrow y'' - \tan x.y' + \sec x e^x.y = 0$
$\Longrightarrow p(x) = -\tan x \Longrightarrow e^{-\int p(x)dx} = e^{\int \tan x dx} = e^{\int \frac{\sin x}{\cos x} dx} = e^{-\int \frac{-\sin x}{\cos x} dx} =$ $e^{-\ln(\cos x)} = e^{\ln(\cos x)^{-1}} = (\cos x)^{-1} = \sec x$

Question-7(Marks-3, Lecture-13)

Given that $y = Ae^x$ is one parameter family of curves satisfying the differential equation: $\frac{dy}{dx} - y = 0$.Then find a member of family satisfying $y(0) = 1$.

Solution:

Given that $y = Ae^x$ under $y(0) = 1$

Put $x = 0$ and $y = 1$ in the given solution.

$\therefore 1 = Ae^0 \Longrightarrow A = 1$, put this in the given solution.

$\therefore y = 1e^x = e^x$ is the required member of the given family.

Question-8(Marks-3, Lecture-14)

Determine whether the following functions are linearly dependent or independent;

$f(x) = a, g(x) = ax$ and $h(x) = ax^2$, $a \in \mathbb{R} - \{0\}$.

Solution:

Taking $W(f, g, h) = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ f'' & g'' & h'' \end{vmatrix} = \begin{vmatrix} a & ax & ax^2 \\ 0 & a & 2ax \\ 0 & 0 & 2a \end{vmatrix} = 2a^3 \neq 0$

$\Longrightarrow$ functions are linearly independent.

Question-9(Marks-3, Lecture-15)

If $y_1(x) = e^{-x}$ be the 1st solution of the differential equation:$y''+2y'+y = 0$, then construct its 2nd solution by using:

$y_2(x) = y_1(x) \int \frac{e^{-\int p(x)dx}}{y_1^2(x)}$, where $p(x)$ is the coefficient of y'.

Solution:

Here $p(x) = 2$

$\Longrightarrow y_2(x) = y_1(x) \int \frac{e^{-\int p(x)dx}}{y_1^2(x)} = e^{-x} \int \frac{e^{-\int 2dx}}{e^{-2x}} dx = e^{-x} \int \frac{e^{-2x}}{e^{-2x}} dx = e^{-x} \int 1dx = xe^{-x} + c$

Question-10 (Marks-5, Lecture-15)

If $y_1(x) = \sin x$ is the 1st solution of the differential equation: $y'' + 9y = 0$, then find its 2nd solution by using

$y_2(x) = y_1 \int \frac{e^{-\int p(x)dx}}{y_1^2} dx$. Also determine their linear dependence or independence.

Solution:

Given that $y'' + 9y = 0 \Longrightarrow p(x) = 0$

$\therefore y_2(x) = y_1 \int \frac{e^{-\int p(x)dx}}{y_1^2} dx = \sin x \int \frac{e^{-\int (0)dx}}{\sin^2 x} dx = \sin x \int \frac{e^c}{\sin^2 x} dx = \sin x \int A \csc^2 x dx$, where $A = e^c \neq 0$

$\Longrightarrow y_2(x) = A \sin x \int \csc^2 x dx = -A \sin x \cot x = -A \cos x$ i.e. the required 2nd solution.

Now for linear independence:

$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \sin x & -A \cos x \\ \cos x & A \sin x \end{vmatrix} = A \sin^2 x + A \cos^2 x = A = e^c \neq 0$

$\Longrightarrow \{y_1, y_2\}$ is linearly independent.

Question-11 (Marks-5, Lecture-14)

Solve the differential equation: $y'' + ay = 0, a > 0$ subject to the boundary conditions: $y(0) = 1$ and $y\left(\frac{\pi}{2\sqrt{a}}\right) = -1$.

Solution:

Let $y = e^{mx}$ is the solution of the given differential equation.

$\Longrightarrow \; y' = me^{mx} \Longrightarrow y'' = m^2 e^{mx}$

$\therefore$the given differential equation becomes;

$m^2 e^{mx} + ae^{mx} = 0 \Longrightarrow e^{mx}\left(m^2 + a\right) = 0$

$\Longrightarrow m^2 + a = 0 \quad \because e^{mx} \neq 0$

$\Longrightarrow m = \pm i\sqrt{a}$, where the roots are imaginary.

$\therefore$ the solution is of the form: $y = \alpha \cos\sqrt{a}x + \beta \sin\sqrt{a}x$

Put $y(0) = 1$ and $y\left(\frac{\pi}{2\sqrt{a}}\right) = -1$ in the above;

$1 = \alpha \cos 0 + \beta \sin 0 \Longrightarrow \boxed{\alpha = 1}$

For $y\left(\frac{\pi}{2\sqrt{a}}\right) = -1; \quad -1 = \alpha \cos\left(\sqrt{a}\frac{\pi}{2\sqrt{a}}\right) + \beta \sin\left(\sqrt{a}\frac{\pi}{2\sqrt{a}}\right) \Longrightarrow -1 = \alpha \cos\left(\frac{1}{2}\pi\right) + \beta \sin\left(\frac{1}{2}\pi\right)$

$\Longrightarrow -1 = \alpha(0) + \beta.1 \Longrightarrow \boxed{\beta = -1}$

$\therefore$ the required particular solution: $\boxed{y = \cos\sqrt{a}x - \sin\sqrt{a}x}$

Question-12 (Marks-5, Lecture-13)

The two solutions of the differential equation: $y'' + 2y' + y = 0$ are e^{-x} and xe^{-x},then prove or disprove that $y = Ae^{-x} + Bxe^{-x}$ is its general solution.

Solution:

$y = Ae^{-x} + Bxe^{-x} \Longrightarrow y' = \frac{d}{dx}\left(Ae^{-x} + Bxe^{-x}\right) = -e^{-x}\left(A - B + Bx\right)$

$y'' = \frac{d}{dx}\left(-e^{-x}\left(A - B + Bx\right)\right) = e^{-x}\left(A - 2B + Bx\right)$

$\therefore y'' + 2y' + y$

$= e^{-x}\left(A - 2B + Bx\right) + 2\left(-e^{-x}\left(A - B + Bx\right)\right) + Ae^{-x} + Bxe^{-x}$

$= e^{-x}\left(A - 2B + Bx - 2\left(A - B + Bx\right) + A + Bx\right) = e^{-x}\left(A - 2B + Bx + 2B - 2A - 2Bx + A + Bx\right)$

$= e^{-x}(0) = 0$

$\Longrightarrow y = Ae^{-x} + Bxe^{-x}$ is the general solution of the given differential equation.

(MCQ, Lecture 16, Marks-01)

1)

If the roots of an auxiliary equation associated with the homogeneous differential equation are $\pm 2, 1 \pm 2i$,then its general solution is ——.

$y = Ae^{2x} + Be^{-2x} + e^{2ix}(C\cos x + D\sin x)$

$y = Ae^{2x} + Be^{-2x} + e^{ix}(C\cos 2x + D\sin 2x)$

$y = Ae^{2x} + Be^{-2x} + e^{x}(C\cos 2x + D\sin 2x)$ $(correct)$

$y = Ae^{2x} + Be^{-2x} + e^{2ix}(C\cos x - D\sin x)$

(MCQ, Lecture 17, Marks-01)

2)

If $a\cos x + b\sin x$ is the solution for $y'' + y = 0$, then which of the following would be general form of the particular solution for $y'' + y = 4\cos x - \sin x$?

$y_p = c\cos x + d\cos x$

$y_p = cx\cos x + dx\cos x \quad (correct)$

$y_p = e^x\,(c\cos x + d\cos x)$

$y_p = e^x\,(cx\cos x + dx\cos x)$

(MCQ, Lecture 18, Marks-01)

3)

If the annihilator of $e^{\pi x}$ and $\left(x^2 + 2x + 3\right)$ are $(D - \pi)$ and D^3 respectively, then the annihilator of their linear combination is —-

$(D - \pi) + D^3$

$(D - \pi) - D^3$

$D^3\,(D - \pi)(correct)$

$\frac{(D-\pi)}{D^3}$

(Descriptive, Lecture 16, Marks-02)

4)

If $y = e^{mx}$ is a solution of the differential equation:$\frac{d^2y}{dx^2} + ay = 0\,, a \in \mathbb{R}\,,$ then construct its associated auxiliary equation.

Solution:

$\because y = e^{mx} \Longrightarrow \frac{dy}{dx} = \frac{d}{dx}e^{mx} = me^{mx} \quad \Longrightarrow \frac{d^2y}{dx^2} = \frac{d}{dx}\left(me^{mx}\right) = m^2e^{mx}$

$\therefore \frac{d^2y}{dx^2} + ay = 0 \Longrightarrow m^2e^{mx} + ae^{mx} = 0$

$e^{mx}\left(a + m^2\right) = 0 \Longrightarrow \boxed{m^2 + a = 0} \quad \because e^{mx} \neq 0$

i.e. the required auxiliary equation.

(Descriptive, Lecture 17, Marks-02)

5)

If $y = e^{-\frac{x}{2}}\left(a\cos\left(\frac{\sqrt{3}}{2}x\right) + b\sin\left(\frac{\sqrt{3}}{2}x\right)\right)$ is a complementary solution of the differential equation: $y'' + y = x\sin x$, then what will be the general form of its particular solution?

Solution:

Since $\sin\left(\frac{\sqrt{3}}{2}x\right)$ and $x\sin x$ are linearly independent, so the general form of its particular solution of the given D.E would be of the form:

$y_p = x(c\cos x + d\sin x)$.

(Descriptive, Lecture 18, Marks-02)

6)

If $(D-\pi)^2\,\pi x e^{\pi x} = 0$, $\left(D^2+\pi^2\right)\cos\pi x = 0$ and $D(\pi x) = 0$, then what will be the annihilator of their linear combination?

Solution:

Linear combination of the given functions: $c_1\pi x e^{\pi x} + c_2\cos\pi x + c_3\pi x$

$\Longrightarrow$its annihilator would be the product of annihilators of individual functions as:

$\left((D-\pi)^2\left(D^2+\pi^2\right)D\right)\left(c_1\pi x e^{\pi x} + c_2\cos\pi x + c_3\pi x\right) = 0$

$\therefore D\left(D^2+\pi^2\right)(D-\pi)^2$ is the required annihilator.

(Descriptive, Lecture 16, Marks-03)

7)

If $m^3+am=0$ is an associated auxiliary equation of the differential equation:$\frac{d^3y}{dx^3} + a\frac{dy}{dx} = 0, a\in\mathbb{R}$, then find its general solution.

Solution:

$\because$given that $m^3+am=0 \Longrightarrow m\left(a+m^2\right)=0 \Longrightarrow m=0 \vee m^2+a=0$

$\Longrightarrow m=0 \vee m^2+a=0 \Longrightarrow m=0\pm i\sqrt{a}$

$\therefore$the required general solution: $y = ae^{0x} + e^{0x}\left(b\cos\left(\sqrt{a}x\right) + c\sin\left(\sqrt{a}x\right)\right) = a + b\cos\left(\sqrt{a}x\right) + c\sin\left(\sqrt{a}x\right)$

(Descriptive, Lecture 17, Marks-03)

8)

What would be the general form of a particular solution of the differential equation:$y''+y = 4\cos x - \sin x$?

Solution:

Associated homogenous D.E corresponding to given is: $y''+y=0$.

If $y=e^{mx}$ be its solution, then the auxiliary equation is:$m^2+1=0 \Longrightarrow m=0\pm i$

$\therefore$ the its general solution: $y_c = a\cos x + b\sin x$

$\because$ the input function contains $\sin x$ and $\cos x$ while complementary solution also contain this, which will no more be independent.

$\Longrightarrow$proposed general form of the particular solution is:

$\boxed{y_p = cx\cos x + dx\sin x}$

(Descriptive, Lecture 18, Marks-03)

9)

Determine the annihilator operator of general solution of the differential equation:$y''+\pi y=0$.

Solution:

Given that $y''+\pi y = 0 \Longrightarrow \frac{d^2y}{dx^2} + \pi y = 0 \Longrightarrow \frac{d^2}{dx^2}y + \pi y = 0$

$\Longrightarrow D^2y + \pi y = 0 \qquad \because D^2 \equiv \frac{d^2}{dx^2}$
$\Longrightarrow \left(D^2 + \pi\right) y = 0$
$\Longrightarrow \left(D^2 + \pi\right)$ is the required annihilator the solution $y = f(x)$ of the given differential equation.

(Descriptive, Lecture 16, Marks-05)

10)

If $m^2 + 1 = 0$ is an auxiliary equation corresponding to the differential equation: $\frac{d^2y}{dx^2} + y = 0$, then find its particular solution satisfying the initial conditions: $y(0) = 1$ and $\frac{dy}{dx}|_{x=\pi} = -1$.

Note: $\frac{dy}{dx}|_{x=\pi} = -1$ means first derivative of y at $x = \pi$ is equal to -1.

Solution:

$\because m^2 + 1 = 0$ is given.$\Longrightarrow m = 0 \pm i$.
$\therefore$ the general solution is: $y = e^{0x}\left(a\cos x + b\sin x\right) = a\cos x + b\sin x$——(1)
$\Longrightarrow \frac{dy}{dx} = -a\sin x + b\cos x$——(2)
For given $y(0) = 1, (1) \Longrightarrow 1 = a\cos 0 + b\sin 0 \Longrightarrow a = 1$
and for $\frac{dy}{dx}|_{x=\pi} = -1, (2) \Longrightarrow -1 = -a\sin\left(\pi\right) + b\cos\left(\pi\right) \Longrightarrow b = 1$
$\therefore$ the required particular solution is: $\boxed{y_p = \cos x + \sin x}$

(Descriptive, Lecture 17, Marks-05)

11)

If the complementary solution of following non-homogenous differential equation is $ae^x + be^{2x}$, then determine its particular solution by using Undetermined Coefficients method.

$y'' - 3y' + 2y = 4e^x$.

Solution:

Here input function:$4e^x$ and particular solution also contains e^x.

$\therefore$ by linear independence, the proposed general form of the particular solution is: $y_p = cxe^x$

$\Longrightarrow y_p' = ce^x + cxe^x = ce^x\left(x + 1\right) \Longrightarrow y_p'' = ce^x\left(x + 2\right)$
Now the given: $y'' - 3y' + 2y = 4e^x$
$\Longrightarrow ce^x\left(x + 2\right) - 3ce^x\left(x + 1\right) + 2cxe^x = 4e^x$
$\Longrightarrow 2ce^x + cxe^x - 3ce^x - 3cxe^x + 2cxe^x = 4e^x$
$\Longrightarrow -ce^x = 4e^x \Longrightarrow \boxed{c = -4}$
$\therefore$the required particular solution:$\boxed{y_p = -4xe^x}$

(Descriptive, Lecture 18, Marks-05)

12)

If $L \equiv D^2 - 5D - 6$ is a linear differential operator such that $Ly = 0$,for $y = f(x)$. Then:

i) Construct a differential equation corresponding to L.

ii) Determine the general solution of differential equation in case of *(i)* .

iii) Determine the annihilator operator of the general solution in case of *(ii)* .

Solution:

i) Given that $Ly = 0 \Longrightarrow \left(D^2 - 5D - 6\right) y = 0 \Longrightarrow D^2y - 5Dy - 6y = 0$

$\because \frac{d}{dx} \equiv D$ is a differential linear operator$\Longrightarrow \frac{d^2}{dx^2} \equiv D$

$\therefore \frac{d^2}{dx^2}y - 5\frac{d}{dx}y - 6y = 0 \Longrightarrow \boxed{\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 6y = 0}$ is the required differential equation.

*ii)*Say $y = e^{mx}$ be its solution$\Longrightarrow \frac{dy}{dx} = me^{mx}$ and $\frac{d^2y}{dx^2} = m^2e^{mx}$

$\therefore$D.E$\Longrightarrow m^2e^{mx} - me^{mx} - 6e^{mx} = 0 \Longrightarrow e^{mx}\left(m+2\right)\left(m-3\right) = 0$

$\Longrightarrow \left(m+2\right)\left(m-3\right) = 0 \quad \because e^{mx} \neq 0$

$\Longrightarrow m = -2, 3$

$\therefore$the general solution: $y = ae^{-2x} + be^{3x}$

iii) $\because$Given that $Ly = 0 \Longrightarrow L \equiv D^2 - 5D - 6$ is the required annihilator operator for the general solution.

1-MCQ, Lecture 19, Marks-01

Which of the following would be an annihilator operator for $\pi x \cos\left(\sqrt{2}x\right)$?

$D^2 + 2$

$\left(D^2 + 2\right)^2$ (correct)

$D^2 + 2\pi$

$\left(D^2 + 2\pi\right)^2$

2-MCQ, Lecture 20, Marks-01

For the differential equation $y'' - y = e^{2x}$, we get $y_1 = e^x, y_2 = e^{-x}, W(y_1, y_2) = -2, W_1 = e^{2x}$ and $W_2 = -e^x$.If $u_1' = -\frac{e^{3x}}{2}$, then $u_1 = ----$.

$-\frac{e^{3x}}{2}$

$\frac{e^{3x}}{2}$

$-\frac{e^{3x}}{6}$ (correct)

$\frac{e^{3x}}{6}$

3-MCQ, Lecture 21, Marks-01

The functions $x, 2x$ and $3x$ are the solutions of a differential equation, then these are linearly$-----$.

independent

dependent (correct)

4-MCQ, Lecture 22, Marks-01

Time period for a body performing Simple Harmonic motion, whose position at any point (x, t) is $x(t) = a\cos t + b\sin t$ is——-.

π

$\frac{\pi}{2}$

$\frac{\pi}{4}$

2π (correct)

5-Descriptive, Lecture 19, Marks-02

Determine the annihilator operator of $px\sin(qx)$,where $p, q \in \mathbb{R}$.

Solution:

$\because D^2(px) = pD^2x = p.0 = 0$

and $\left(D^2 + q^2\right)\sin(qx) = D^2\left(\sin(qx)\right) + q^2\sin(qx) = -q^2\sin(qx) + q^2\sin(qx) = 0$

$\therefore$Annihilator of $px\sin(qx) = \left(D^2 + q^2\right)^2$ i.e. $\left(D^2 + q^2\right)^2\left(px\sin(qx)\right) = 0$

6-Descriptive, Lecture 20, Marks-02

For the differential equation:$a_2(x)y' + a_1(x)y + a_0(x) = f(x), a_2(x) \neq 0$, write the formulation for its general solution.

Solution:

$a_2(x)y' + a_1(x)y + a_0(x) = f(x) \Longrightarrow y' + \frac{a_1(x)}{a_2(x)}y + \frac{a_0(x)}{a_2(x)} = \frac{f(x)}{a_2(x)}$

$\Longrightarrow y' + p(x)y + q(x) = g(x)$, where $\frac{a_1(x)}{a_2(x)} = p(x), \frac{a_0(x)}{a_2(x)} = q(x)$ and $\frac{f(x)}{a_2(x)} = g(x)$.

Now its general solution will be given by;

$$\boxed{y = \underbrace{e^{-\int p(x)dx}\int e^{\int p(x)dx} g(x)dx}_{\text{Particular Solution}} + \underbrace{ce^{-\int p(x)dx}}_{\text{Complementary Solution}}}$$

7-Descriptive, Lecture 21, Marks-02

If we have a differential equation: $\frac{d^4y}{dx^4} + ky = x + e^{-x}$, then commemorate the idea of variation of parameters, write down the third column of W_3.

Solution:

$\because$ the given DE is of order 4

$\Longrightarrow$ there will be 4 linearly independent solutions.

$\Longrightarrow$Wronskian would be of order 4×4.

$\Longrightarrow$3rd column: $\begin{pmatrix} 0 \\ 0 \\ 0 \\ x + e^{-x} \end{pmatrix}$

8-Descriptive, Lecture 22, Marks-02

If the position x at any time t, for a body performing simple Harmonic motion is; $x(t) = \frac{\cos t}{2} + \frac{\sqrt{3}\sin t}{2}$, then find its amplitude of the vibration.

Solution:

Comparing the given: $x(t) = \frac{\cos t}{2} + \frac{\sqrt{3}\sin t}{2} = \frac{1}{2}\cos t + \frac{\sqrt{3}}{2}\sin t$, with the general form of SHM:

$x(t) = c_1 \cos t + c_2 \sin t \Longrightarrow c_1 = \frac{1}{2}$ and $c_2 = \frac{\sqrt{3}}{2}$

$\therefore$Amplitude$= \sqrt{c_1^2 + c_2^2} = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)} = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1$

9-Descriptive, Lecture 19, Marks-03

If the complementary solution of $y'' + \pi y = \cos \pi x$ is $y_c = a\cos\sqrt{\pi}x + b\sin\sqrt{\pi}x$, then find the general solution of the higher-order homogeneous differential equation obtained by multiplying the given differential equation with the annihilator operator $D^2 + \pi^2$.

Solution:

Operator of the given DE is:

$\left(D^2 + \pi\right) y = \cos \pi x$

According to given condition;

$\left(D^2 + \pi^2\right)\left(D^2 + \pi\right) y = \left(D^2 + \pi^2\right)\cos \pi x$

$\Longrightarrow \left(D^2 + \pi^2\right)\left(D^2 + \pi\right) y = 0 \qquad \because \begin{cases} \text{Annihilator of } \cos\pi x \text{ is } D^2 + \pi^2 \\ \text{i.e. } \left(D^2 + \pi^2\right)\cos \pi x = 0 \end{cases}$

$\therefore$Associated auxiliary equation to the above Homogeneous DE is;

$\left(m^2 + \pi^2\right)\left(m^2 + \pi\right) = 0 \Longrightarrow m = \pm i\sqrt{\pi}, \pm i\pi$

$\therefore$ the general sol: $\boxed{y = k_1 \cos\sqrt{\pi}x + k_2 \sin\sqrt{\pi}x + k_3 \cos \pi x + k_4 \sin \pi x}$

10-Descriptive, Lecture 20, Marks-03

Find the auxiliary equation associated to the differential equation: $y'' + ay = \sec x$, for $a > 0$ and $a < 0$.

Solution:

For $a > 0$, if $y = e^{mx}$ be its solution, then associated homogeneous DE is;
$m^2e^{mx} + ae^{mx} = 0 \Longrightarrow e^{mx}\left(a + m^2\right) = 0 \Longrightarrow \boxed{m^2 + a = 0} \quad \because e^{mx} \neq 0$

For $a < 0 \exists b > 0$ in $\mathbb{R}$ such that $a = -b$, and if $y = e^{mx}$ be its solution, then associated homogeneous DE is;
$m^2e^{mx} - be^{mx} = 0 \Longrightarrow e^{mx}\left(-b + m^2\right) = 0 \Longrightarrow \boxed{m^2 - b = 0} \quad \because e^{mx} \neq 0$

11-Descriptive, Lecture 21, Marks-03

Determine W by applying the technique of the variation of parameters on the following differential equation:
$\frac{d^3y}{dx^3} + \pi\frac{dy}{dx} = \tan 2x$

Solution:

If $y = e^{mx}$ is the proposed solution of the associated homogeneous DE of the given, then the auxiliary eq: is
$m^3 + \pi m = 0 \Longrightarrow m\left(\pi + m^2\right) = 0 \Longrightarrow m = 0, 0 \pm i\sqrt{\pi}$
$\Longrightarrow y_c = ae^0 + e^0\left(b\cos\sqrt{\pi}x + d\sin\sqrt{\pi}x\right) = a.1 + b\cos\sqrt{\pi}x + d\sin\sqrt{\pi}x$
$\Longrightarrow$fundamental set$= \{1, \cos\sqrt{\pi}x, \sin\sqrt{\pi}x\}$

$$\Longrightarrow W\left(1, \cos\sqrt{\pi}x, \sin\sqrt{\pi}x\right) = \begin{vmatrix} 1 & \cos\sqrt{\pi}x & \sin\sqrt{\pi}x \\ 0 & -\sqrt{\pi}\sin\sqrt{\pi}x & \sqrt{\pi}\cos\sqrt{\pi}x \\ 0 & -\pi\cos\sqrt{\pi}x & -\pi\sin\sqrt{\pi}x \end{vmatrix} = \pi\sqrt{\pi}\sin^2\sqrt{\pi}x +$$
$\pi\sqrt{\pi}\cos^2\sqrt{\pi}x = \pi\sqrt{\pi}$

12-Descriptive, Lecture 22, Marks-03

Determine the equation of the simple harmonic motion for a mass weighing w lbs, which stretches a spring to a inches from an initial position.

Solution:

$\because$Weight$= F = W = w$lbs$\Longrightarrow mg = w \Longrightarrow m = \frac{w}{g} = \frac{w}{32}$
distance from Initial position$= s = a$ inches$= \frac{a}{12}$ft
$\therefore$By Hook's law, $F = ks \Longrightarrow \frac{w}{32} = k\frac{a}{12} \Longrightarrow k = \frac{3w}{8a}$

Now the eq: of SHM: $m\frac{d^2x}{dt^2} = -kx \Longrightarrow \frac{w}{32}\frac{d^2x}{dt^2} = -\frac{3w}{8a}x \Longrightarrow \boxed{\frac{d^2x}{dt^2} + \frac{12}{a}x = 0}$ is the required equation.

13-Descriptive, Lecture 19, Marks-05

If the complementary solution of $y'' + 3y' + 2y = e^x\sin(2x + 3)$ is $y_c = ae^{-x} + be^{-2x}$, then determine the form of its particular solution by applying **only** the concept of annihilator operator approach.

Solution:

Operator form of the given DE is;
$\left(D^2 + 3D + 2\right)y = e^x\sin(2x + 3)$
And the Annihilator of $e^{1x}\sin(2x+3) = D^2 + 1^2 + 2^2 - 2(1)D \equiv D^2 - 2D + 5$
i.e. $\left(D^2 - 2D + 5\right)e^x\sin(2x + 3) = 0$
$\therefore$Pre-operating by $\left(D^2 - 2D + 5\right)$ on both sides of given D.E
$\Longrightarrow \left(D^2 - 2D + 5\right)\left(D^2 + 3D + 2\right)y = \left(D^2 - 2D + 5\right)e^x\sin(2x + 3)$
$\Longrightarrow \left(D^2 - 2D + 5\right)\left(D^2 + 3D + 2\right)y = 0$
$\Longrightarrow$associated auxiliary eq: of above Homogeneous eq: is;
$\Longrightarrow \left(m^2 - 2m + 5\right)\left(m^2 + 3m + 2\right) = 0$
$\Longrightarrow m = -2, -1$ and $m^2 - 2m + 5 = 0 \Longrightarrow m = 1 + 2i, 1 - 2i$

$\therefore$ the general sol: $\boxed{y = Ae^{-x} + Be^{-2x} + e^{x}(D\cos 2x + E\sin 2x)}$

14-Descriptive, Lecture 20, Marks-05

Find the wronskian for the complementary solution of the differential equation:$y''+ ay = \sec x$,for $a > 0$; using the variation of parameter.

Solution:

For $a > 0$, if $y = e^{mx}$ be its solution, then associated homogeneous DE:
$y'' + ay = 0 \Longrightarrow m^2 e^{mx} + ae^{mx} = 0 \Longrightarrow e^{mx}\left(a + m^2\right) = 0$
$\Longrightarrow m^2 + a = 0 \ \because e^{mx} \neq 0$
$\Longrightarrow m = 0 \pm i\sqrt{a}$
$\Longrightarrow$the complementary solution:$y_c = A\cos\left(\sqrt{a}x\right) + B\sin\left(\sqrt{a}x\right)$
$\Longrightarrow$Fundamental Set$= \{y_1, y_2\} = \left\{\cos\left(\sqrt{a}x\right), \sin\left(\sqrt{a}x\right)\right\}$
$\Longrightarrow$Wronskian$= W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos\left(\sqrt{a}x\right) & \sin\left(\sqrt{a}x\right) \\ -\sqrt{a}\sin\left(\sqrt{a}x\right) & \sqrt{a}\cos\left(\sqrt{a}x\right) \end{vmatrix} = \sqrt{a}\cos^2\left(\sqrt{a}x\right) + \sqrt{a}\sin^2\left(\sqrt{a}x\right) = \sqrt{a}\left(\cos^2\sqrt{a}x + \sin^2\sqrt{a}x\right) = \sqrt{a}$

15-Descriptive, Lecture 21, Marks-05

Solve the differential equation:$\frac{d^2y}{dx^2} + y = \tan^3 x$ by variation of parameters.Where:

$y_c = ay_1 + by_2 = a\cos x + b\sin x, W\left(y_1, y_2\right) = 1, W_1 = -\sin x\tan^3 x$ and $W_2 = \cos x\tan^3 x$.

Solution:

$\because$we have given the complementary solution and just to find the particular one, which is given by:

$y_p = u_1(x)y_1(x) + u_2(x)y_2(x)$,where $u_1 = \int \frac{W}{W_1}dx$ and $u_2 = \int \frac{W_2}{W}dx$.

Now $u_1 = \int \frac{W_1}{W}dx = \int \frac{-\sin x\tan^3 x}{1}dx = -\int \frac{\sin^4 x}{\cos^3 x}dx$
and $u_2 = \int \frac{W_2}{W}dx = \int \frac{\cos x\tan^3 x}{1}dx = \int \frac{\sin^3 x}{\cos^2 x}dx$

Here both the integrals are not of fundamental nature and will remain as such.

$\therefore$ the general sol:$y = y_c + y_p = a\cos x + b\sin x + u_1(x)y_1(x) + u_2(x)y_2(x) = a\cos x + b\sin x + \cos x\left(-\int \frac{\sin^4 x}{\cos^3 x}dx\right) + \sin x\left(\int \frac{\sin^3 x}{\cos^2 x}dx\right)$

$\boxed{y = a\cos x + b\sin x - \cos x\int \frac{\sin^4 x}{\cos^3 x}dx + \sin x\int \frac{\sin^3 x}{\cos^2 x}dx}$

16-Descriptive, Lecture 22, Marks-05

Express the solution:$x(t) = a\cos\omega t - b\sin\omega t$ of Simple Harmonic Motion(SHM) in the form of $x(t) = A\cos(\omega t + \phi)$.

Solution:

Firstly we look for to find A and ϕ
$\because x(t) = A\cos(\omega t - \phi)$ is the required form;
$\therefore$ {by using $\cos\left(\alpha - \beta\right) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$
$\Longrightarrow x(t) = A\left(\cos t\omega\cos\phi + \sin t\omega\sin\phi\right) = (A\cos\phi)\cos t\omega + (A\sin\phi)\sin\omega t\,(A\sin\phi) = (A\cos\phi)\cos t\omega + A\sin\phi\sin\omega t$

Comparing this with given:$x(t) = a\cos\omega t - b\sin\omega t$

$\Longrightarrow a = A\cos\phi$ and $b = (-A\sin\phi)$

Squaring and Adding;

$a^2+b^2 = (A\cos\phi)^2+(A\sin\phi)^2 = A^2\cos^2\phi+A^2\sin^2\phi = A^2\left(\cos^2\phi+\sin^2\phi\right) = A^2.1 = A^2$

$\Longrightarrow A^2 = a^2+b^2 \Longrightarrow A = \sqrt{a^2+b^2}$

Again $a = A\cos\phi \Longrightarrow \frac{a}{A} = \cos\phi$ and $\frac{b}{A} = \sin\phi$

$\Longrightarrow \tan\phi = \frac{\sin\phi}{\cos\phi} = \frac{\frac{b}{A}}{\frac{a}{A}} \Longrightarrow \phi = \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{b}{a}\right)$

Now from the given:$x(t) = a\cos\omega t - b\sin\omega t = A\left(\frac{a}{A}\cos\omega t - \frac{b}{A}\sin\omega t\right)$ (*Note this step*)

$\Longrightarrow x(t) = A\left(\frac{a}{A}\cos\omega t - \frac{b}{A}\sin\omega t\right) = A\left(\cos\phi\cos\omega t - \sin\phi\sin\omega t\right) = A\left(\cos\omega t\cos\phi - \sin\omega t\sin\phi\right)$

$\because \{\cos\alpha\cos\beta - \sin\alpha\sin\beta = \cos(\alpha+\beta)$

$\therefore \boxed{x(t) = A\cos(\omega t+\phi) = \sqrt{a^2+b^2}\cos\left(\omega t+\tan^{-1}\left(\frac{b}{a}\right)\right)}$ i.e. the required form.